Why LLMs Suddenly 'Understand': A Protocol-Compiled Regime-Transition Model Integrating Fourier-Mode Selection, Collapse-Without-Alignment Macro Coherence, SMFT Projection, and the PORE Ξ-Stack

0. Reader Contract, Claim Level, and Non-Claims

0.1 Aim (what you will get if you read this paper)

This paper proposes a **portable explanation template** for why an LLM can appear to **suddenly "understand"** after seeing many examples. The template treats sudden understanding as a **protocol-relative regime transition** inside a training loop, rather than a mystical capability jump.

The integration uses four ingredients:

Mechanistic toy laboratory: modular addition dynamics (Fourier features, lottery-ticket mode selection, grokking staging).

Macro stability principle: Collapse Without Alignment (CWA): macro predictability can hold under micro heterogeneity via additive projection.

Engineering discipline: PORE / Minimal Intrinsic Triple: protocol-first compilation into stable effective coordinates with explicit operator channels.

Generic collapse grammar: SMFT-style "projection/selection" as an internal step (introduced later, operationally, not metaphysically). (No new claims are required here; we only use the projection *pattern*.)

0.2 Claim level (what is asserted)

We claim:

C0 (Operational claim): "Sudden understanding" can be modeled as crossing a **critical surface** in a small set of **compiled order parameters** $\Xi(t)$ under a declared protocol P.
C1 (Explanatory claim): A minimal, reusable explanation exists that decomposes the event into:
(i) **micro-level mode selection** (collapse-by-competition), plus
(ii) **macro-level noise cancellation** (CWA), plus

(iii) **protocol-fixed compilation and falsification gates** (PORE discipline).

We do **not** claim that the modular-addition mechanism (Fourier features etc.) is literally present in all LLM tasks; it is used as a *clean template* showing how such transitions can be mechanistically real.

0.3 Non-claims (what we explicitly do NOT claim)

NC1 (No new ontology): we do not claim a new physical theory of reality.

NC2 (No single privileged internal basis): we do not claim Fourier is the universal basis for LLM understanding; Fourier is the *toy-case basis forced by symmetry*.

NC3 (No guarantee): we do not claim sudden understanding will always occur, or at a predictable step count, without specifying protocol P.

NC4 (No interpretability shortcut): we do not claim we can "read understanding" directly from hidden weights in all LLMs; we instead insist on **compiled observables** and **operator tests**.

0.4 How to use this paper (recommended workflow)

Declare your training loop as a protocol P.

Choose a measurement/compression map h (what you can log).

Compute a small set of order parameters $\hat{\Xi}(t)$ (proxies).

Detect whether a sharp generalization jump corresponds to a **regime boundary crossing** and which operator channel caused it.

If diagnostics fail, repair protocol / probes instead of patching narratives.

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1. Problem Statement: "Sudden Understanding" as a Measurable Event

1.1 The phenomenon

Empirically, many learning systems display a pattern:

long period of mediocre out-of-sample behavior (or "memorization"), then

a relatively sharp transition into strong generalization ("it suddenly gets it").

In modular arithmetic, this is studied as **grokking**, where test performance improves long after training loss has already collapsed.

We use grokking only as a *canonical demonstration* that the phenomenon can be real and mechanistically analyzable.

1.2 What we mean by “sudden”

We need a definition that is operational under a protocol.

Let G(t) be a generalization score (e.g., held-out accuracy, loss gap, or task-dependent metric) computed from the protocol-bound log.

We define the event:

(1.1) SuddenUnderstanding(P) := 1[∃t: G(t) crosses a threshold with steep slope under fixed P]

Where “steep slope” is also protocol-dependent (window size, smoothing, timebase). This is intentional: **PORE forbids undefined objects without protocol.**

1.3 The question we actually want to answer

Not “why is the model smart,” but:

Q1: What minimal internal dynamic can produce a delayed but sharp improvement?

Q2: Why does it require many examples / long training time?

Q3: Why can the transition look sudden even when internal change is gradual?

Q4: How can we test this explanation without reading the entire model?

The integrated answer preview (no full theory yet):

Micro: mode selection can be winner-take-most (collapse-like), so a tiny advantage can amplify slowly, then dominate rapidly.

Macro: even if micro remains heterogeneous, an additive projection can suddenly become stable when SNR crosses a threshold (CWA).

Engineering: we must compile and test these claims under a declared protocol with operator channels (PORE).

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2. The PORE Compilation Frame: From Training Reality to Reportable Dynamics

2.1 Protocol-first stance (why we start here)

PORE’s central move is:

If you cannot state your protocol, you cannot state your object.

That is, an “object” is not metaphysically given; it is an **effective regularity under a declared boundary, sampling timebase, observation map, and admissible interventions**.

2.2 The protocol package P

We define:

(2.1) $P = (B, \Delta, h, u)$

where:

B (boundary): what is treated as "inside the loop-object" vs outside/exogenous.

Δ (timebase): what one "tick" means (step/epoch/window cadence).

h (observation map): measurement/compression operator mapping microstate → logged observable.

u (operator channels): admissible intervention knobs, treated as first-class and reportable.

2.3 Logging discipline (what the "compiler" is allowed to see)

Under fixed P, we do not assume omniscience. We only use the protocol-bound log:

(2.2) $z[n] = h(x(t_o + n\Delta))$

Interpretation:

x(t) is the full internal training state (weights, optimizer state, data ordering state, regularizers, etc.).

z[n] is what we *actually* log and commit to using for claims.

This is the anti-handwaving constraint: any later "explanation" must be reconstructible from z[n] under P, or else it is outside-scope.

2.4 Minimal operator grammar (how we talk about interventions)

PORE standardizes:

(2.3) $u \in \{\text{Pump}, \text{Probe}, \text{Switch}, \text{Couple}\}$

Operational meanings (brief, expanded later):

Pump: reshape resources / basin depth (energize, stabilize, replenish).

Probe: measure/estimate without secretly changing the system.

Switch: trigger or gate regime transitions (e.g., regularization schedules, curriculum shifts).

Couple: change interaction structure / coherence (connect subsystems, enforce symmetry, share representations).

This vocabulary will later let us describe "sudden understanding" as an **operator-driven crossing** in compiled coordinates, instead of an anecdote.

2.5 Why this matters for "sudden understanding"

Without protocol discipline, “it understood” is not a stable object:

Change train/test split → different “understanding” time.

Change logging cadence Δ → different perceived suddenness.

Change what you measure (h) → different inferred mechanism.

PORE’s stance is: those are not nuisances; they are **part of the object definition**.

If you want me to continue, the natural next chunk is **Section 3 (Ξ order parameters)**, where we introduce Ξ(t) = (ρ(t), γ(t), τ(t)) as the minimal regime coordinates and define the “critical surface” concept precisely under P.

3. Minimal Intrinsic Triple as Order Parameters for Learning

This section introduces a **three-coordinate order-parameter system** for learning transitions. The goal is not to “explain everything,” but to define a *minimal* state summary that is (i) **operationally interpretable**, (ii) **protocol-compilable**, and (iii) **stable enough** to support regime-transition claims.

The construction follows the Minimal Intrinsic Triple / PORE discipline: **Ξ is a control-coordinate summary, not an ontological statement.**

3.1 Definition (Ξ as learning order parameters)

We define the intrinsic triple as a time-indexed vector of effective scalars:

(3.1) Ξ(t) := (ρ(t), γ(t), τ(t))

Each component is defined **by functional role**, not by a single universal formula. In the original formulation, Ξ is introduced as a “minimal control coordinate system” for open, dissipative systems, explicitly avoiding ontological claims.

For learning systems, we interpret Ξ(t) as an **order-parameter coordinate system**: regions in Ξ-space correspond to qualitatively stable regimes (memorization-like, transition-like, generalization-like), while “sudden understanding” corresponds to crossing a protocol-relative boundary in this space.

3.2 Operational meanings in learning systems (ρ, γ, τ)

3.2.1 ρ — representational mass / occupancy of structure

In the intrinsic triple, ρ is an effective occupancy/density scale: it distinguishes “dilute” from “loaded/concentrated” regimes.

For learning, we use:

ρ(t) = “how much of the model’s capacity is *actually concentrated* into reusable structure,”
not merely how large weights are.

Examples of operational readings:

concentration of representation into a small number of stable directions / modes,

concentration of predictive power into shared circuits rather than per-example quirks,

compression-like signatures (structure becomes denser in a small basis).

3.2.2 γ — coupling / coherence / domain-lock strength

In the intrinsic triple, γ summarizes “how strongly the system is confined, constrained, or symmetry-locked” and separates weakly constrained diffusion from strongly locked trapping.

For learning, we use:

γ(t) = “how strongly subsystems mutually reinforce a shared basis,”
i.e., the degree of **coherence** among interacting parts (layers/heads/features) under the protocol.

Operational readings:

strength of cross-component consistency (features align into a coherent algorithm),

strength of “domain lock” to an invariant structure induced by the task and training setup,

degree to which the learned representation resists being washed out by noise/perturbations.

3.2.3 τ — agitation / dephasing / effective timescale separation

In the intrinsic triple, τ is “effective noise/dephasing/agitation” that smears structure and destroys coherence.

For learning, we use:

τ(t) = “how fast coherence is degraded or smeared,” or equivalently, how hard it is for structure to persist.

Operational readings:

the effective noise level in parameter/feature evolution (stochasticity, interference, churn),

the “dephasing” between competing hypotheses/circuits,

the degree of timescale separation between fast fitting and slow cleanup (a key grokking signature later).

3.3 Dynamics: Ξ as a compressed trajectory under protocol P

We treat Ξ(t) as the **coarse-grained induced trajectory** of a much larger internal state x(t),

compiled through a declared protocol and observation map. The intrinsic triple paper explicitly frames Ξ as a compressed coordinate induced by more detailed evolution (fields/microstates), with coarse-graining made explicit.

Accordingly, we model Ξ's evolution as an effective dynamical system:

(3.2) $d\Xi/dt = F(\Xi; P) + \varepsilon(t)$

$F(\cdot;P)$ is protocol-relative effective flow (depends on boundary B, timebase Δ, observation map h, and admissible operator channels u).

$\varepsilon(t)$ is the residual (untracked degrees of freedom, estimation error, omitted variables).

Key discipline: Ξ is only accepted as a "coordinate" if it is **identifiable and stable** under the declared protocol. PORE explicitly treats compiled coordinates as existing *only where gates pass* (proxy stability, probe backreaction checks, etc.).

3.4 Identifiability: many proxy sets can represent the same Ξ

A central point of the intrinsic triple framework is that ρ, γ, τ are **role-defined**, not estimator-defined. Therefore, in practice, there are multiple legitimate proxy families for each coordinate, and identifiability becomes an engineering problem:

Are different proxies monotone-consistent with the intended role?

Do they remain stable under small protocol-preserving perturbations?

Do they support reproducible regime segmentation?

This is not hypothetical—CAFT explicitly emphasizes diagnostics and "measurement playbooks" for these coordinates, along with tests (e.g., permutation tests) to check whether the macro behavior is consistent with the assumed aggregation model (CWA).

3.4.1 Proxy families (conceptual; concrete recipes come later)

Below are **proxy families**, not final prescriptions. They illustrate identifiability freedom while keeping the intended role intact.

$\hat{\rho}$ proxy families (structure concentration)

spectral concentration: "how much energy is concentrated in top components"

sparsity of dominant features/circuits (e.g., IPR-like measures in a chosen basis)

compression proxies: description-length / effective rank trends

$\hat{\gamma}$ proxy families (coherence / coupling)

cross-module agreement: "do multiple parts compute compatible intermediate claims?"

redundancy measures: “does the same decision appear via multiple pathways?”

stability under perturbations: “does the learned mapping persist under small internal rewiring/ noise?”

$\hat{\tau}$ proxy families (agitation / timescale separation)

volatility of feature directions over time (how quickly the representation rotates/churns)

separation between “fit time” and “generalize time” (grokking delay as τ-like signature)

probe-induced disturbance scores when probes are non-exchangeable (used only if gates demand)

3.5 Minimal requirements (what must be true for Ξ to be meaningful)

The intrinsic triple framework states a minimal requirement: each coordinate must be **monotone with respect to its intended role**.

We adopt the same minimal requirement in learning:

increasing ρ ⇒ “more concentrated reusable structure”

increasing γ ⇒ “stronger coherence / tighter algorithmic lock-in”

increasing τ ⇒ “more agitation / more dephasing / weaker persistence of structure”

Protocol rule: if a proposed estimator violates monotonicity under obvious counterfactuals (e.g., increasing decay increases “$\hat{\rho}$”), it is rejected as a proxy.

Gate rule (PORE): even monotone proxies are not accepted unless they are stable and identifiable under the declared protocol. If Gate 1 (proxy stability) or Gate 3 (probe backreaction / non-exchangeability) fails, you do not patch the story—you revise P, h, or the estimator family.

3.6 Why Ξ is the right “slot” for sudden understanding

The working hypothesis of this paper is:

“Sudden understanding” is not a mysterious new faculty; it is a **crossing event** in a low-dimensional order-parameter space.

The event looks sudden because (i) selection dynamics can be winner-take-most at micro level, and (ii) macro stability can jump when cancellation/SNR crosses threshold under additive aggregation (CWA).

Section 4 will introduce modular addition as a canonical laboratory where these statements can be made unusually explicit, and where “Fourier feature concentration,” “phase coordination,” and “grokking delay” become concrete stand-ins for ρ, γ, τ.

4. Canonical Toy Universe: Modular Addition as a Clean “Mechanistic Laboratory”

This section motivates **modular addition** as a "toy universe" where (i) the task symmetry strongly constrains what a solution can look like, (ii) learned parameters become **directly readable** under a natural change of basis, and (iii) the system exhibits a **delayed → sudden** generalization transition (grokking) that can be dissected end-to-end.

4.1 Why modular addition is special: symmetry ⇒ Fourier basis inevitability

The task is:

(4.1) $z = (x + y) \bmod p$

This is not just "a simple dataset"; it is a function on the cyclic group Z_p with a built-in translation structure. The paper emphasizes that prior work has shown models solve this by learning a **Fourier feature representation**, effectively embedding inputs on a circle so that addition becomes geometric rotation—i.e., the diagonal/eigen coordinate system is Fourier-like.

Key laboratory advantage: because the symmetry selects a privileged basis, we can define an observation map h that is unusually "lossless" for mechanism:

(4.2) $h_{DFT}: (\theta, \xi) \mapsto \{\text{Fourier magnitudes and phases per neuron}\}$

The paper makes this the central analytical technique: apply Discrete Fourier Transform (DFT) to the input/output weight vectors of each neuron and read off a small set of parameters (dominant frequency, magnitude, phase).

PORE interpretation (bridging to Sections 1–3): modular addition supplies a rare case where the Ξ-compilation map h is "obvious" and produces stable, low-dimensional invariants—exactly what we need to study regime transitions without handwaving.

4.2 Empirical invariants: what repeatedly shows up across runs

The paper's Section 3 reports several "invariants" that appear consistently under standard training. These are the backbone of why modular addition functions as a mechanistic lab.

4.2.1 Invariant A: single-frequency Fourier feature per neuron

Observation 1 states that for each neuron m there exists a learned frequency $\varphi(m)$, magnitudes α_m, β_m, and phases ϕ_m, ψ_m such that the weight vectors are well-approximated by cosine waves:

(4.3) $\theta_m[j] = \alpha_m \cdot \cos(\omega_{\varphi(m)} \cdot j + \phi_m)$, $\xi_m[j] = \beta_m \cdot \cos(\omega_{\varphi(m)} \cdot j + \psi_m)$ for all (m,j)

with $\omega_k = 2\pi k/p$.

Interpretation: in the Fourier domain, each neuron becomes **sparse**—one dominant active frequency.

4.2.2 Invariant B: layer-wise phase coupling (“doubled phase”)

Observation 2 reports a precise phase relation:

(4.4) $(2\phi_m - \psi_m) \bmod 2\pi = 0$

Empirically, the pairs $(2\phi_m, \psi_m)$ lie close to the line y=x, indicating that input and output weights couple tightly in Fourier space.

4.2.3 Invariant C: population-level diversification (frequency coverage + phase symmetry)

When width M is large, learned neurons become “fully diversified” across frequencies and phases. The paper formalizes this as **Definition 4.1 (Full Diversification)**, consisting of:

balanced frequency coverage: each frequency k has exactly N neurons,

homogeneous scaling: $\alpha_m \cdot \beta_m^2$ is constant (a),

high-order phase symmetry within each frequency group.

This is not a cosmetic observation: it is the bridge from “local Fourier features” to a **global algorithm**.

4.3 Key extraction 1: the feature representation template (cosine mode + phase)

For reuse later (in LLM contexts where the basis is not literally Fourier), the reusable pattern is not “cosines,” but **mode decomposition**:

each micro-unit develops a dominant mode label (here: $\varphi(m)$),

each unit has an amplitude-like scale (here: α_m, β_m),

each unit has a phase/offset-like degree of freedom (here: ϕ_m, ψ_m),

training couples subparts of the unit so the “readout phase” becomes a predictable transform of the “encoding phase” (here: doubled phase).

In modular addition, the template is explicit and measurable via DFT.

4.4 Key extraction 2: population-level cancellation is the pathway to correctness

The central mechanistic result is that **individual neurons are noisy**, yet the population can implement the correct rule via **majority-voting / cancellation** in Fourier space when diversification holds.

4.4.1 Full diversification ⇒ a “flawed indicator” with a strong signal peak

Under (i) the parametrization (4.3), (ii) full diversification (Definition 4.1), and (iii) phase

alignment (4.4), the paper proves **Proposition 4.2**, deriving a closed-form expression for the output logit at class j:

(4.5) $f(x,y)[j] = aN/2 \cdot \{ -1 + (p/2) \cdot 1[(x+y) \bmod p = j] \} + (p/4) \cdot \Sigma_{z \in \{x,y\}} 1[(2z) \bmod p = j]$

The first bracketed term is the **signal**: it boosts the correct class $j=(x+y) \bmod p$.

The last term is structured **noise**: spurious boosts at $j=2x$ and $j=2y$.

The paper then shows that with suitable scaling, softmax of f approximates the empirical one-hot target distribution up to ε (in a norm they specify).

4.4.2 Why this matters for "sudden understanding"

This gives a concrete, non-handwavy mechanism for a "snap":

The system does not need each unit to be perfect.

It needs **enough diversified units** so the aggregate produces a dominant, stable signal peak.

This is exactly the kind of macro stabilization later captured abstractly by CWA-style additive coherence: heterogeneous micro states can yield stable macro output via aggregation and symmetry.

4.5 What is portable vs task-specific

4.5.1 Portable mechanisms (expected to generalize in form)

P1. Privileged basis emerges from invariances.
Modular addition makes the privileged basis explicit (Fourier). In broader tasks, the basis may be "feature subspaces," "circuits," or "latent factors," but the pattern "symmetry/invariance ⇒ diagonal coordinates" is reusable.

P2. Winner-take-most mode selection under competition.
The paper's lottery-ticket story (developed later) treats feature emergence as competitive amplification of one component inside a unit.

P3. Macro correctness via redundancy + cancellation.
The fully diversified population implements a robust decision rule even though each neuron is noisy.

P4. Regime staging under competing forces.
Grokking is explained as a multi-phase process governed by loss minimization vs weight decay, with measurable progress variables (phase difference, sparsity via IPR, norm).

4.5.2 Task-specific artifacts (do NOT over-transfer literally)

T1. Exact cosine parametrization and doubled-phase law.

These depend on the modular group structure and the two-layer setup where DFT is the natural lens.

T2. Specific "noise terms" 1[2x=j] and 1[2y=j].
These are algebraic consequences of the particular trigonometric identity path in modular addition; other tasks will have different structured failure modes.

T3. Exact frequency balance assumption in full diversification.
Definition 4.1 includes an idealized "exact N per frequency" balance; the paper notes it is approximate under random init and studies ablations that show full frequency/phase diversity is crucial.

4.6 How this plugs into Ξ(t) = (ρ, γ, τ) from Section 3 (preview, not full proxy recipe yet)

Modular addition supplies unusually clean stand-ins for the intrinsic triple roles:

ρ-like (structure concentration): sparsity of Fourier coefficients per neuron (tracked by IPR).

γ-like (coherence/coupling): phase coupling level measured by $|\sin(D_m)|$ *with* $D_m := (2\phi_m - \psi_m) \bmod 2\pi$.

τ-like (agitation/dephasing): persistence of "perturbed Fourier solution" noise across frequencies, and the slow cleanup timescale when weight decay dominates late.

We do not commit to these as universal estimators; we treat them as a **laboratory calibration** that demonstrates what it looks like when Ξ coordinates are well-defined and regime transitions are measurable.

Next section (5) will use this laboratory to extract the microdynamics law: mode competition as a collapse process (lottery-ticket selection) and the minimal amplitude–phase equations that later become the generic "collapse" component of sudden understanding.

5. Microdynamics: Mode Competition as Collapse (Lottery Ticket Selection)

This section extracts the **minimal microdynamic template** behind "sudden understanding" from the modular-addition laboratory: **many candidate modes coexist early, then training amplifies one mode until it dominates**. The modular-addition paper formalizes this as a *lottery-ticket mechanism* driven by coupled **amplitude growth** and **phase-mismatch relaxation**.

The key point for our integrated model is not "Fourier" per se, but the **generic dynamical shape**:

a set of competing hypotheses/modes,

each has a growth rate that depends on an internal coherence variable (phase mismatch),

coherence improves growth, growth accelerates coherence (positive feedback),

the system exhibits **winner-take-most collapse** onto a single dominant mode.

5.1 Mode coordinates: amplitude A_k(t) and mismatch D_k(t)

In modular addition, each neuron's weights become dominated by a **single Fourier frequency** with a phase, and training dynamics can be written in that basis.

Abstracting away from Fourier specifics, we model each candidate mode k (a "feature hypothesis") inside a unit by:

$A_k(t) \geq 0$: mode amplitude (how much representational mass sits in mode k)

$D_k(t) \in (-\pi, \pi]$: mismatch (a phase-like internal inconsistency variable)

In the modular-addition paper, D_k corresponds to a phase relation between input and output weights (a "doubled phase" mismatch), and they show it is driven toward zero under training.

5.2 The minimal coupled flow (alignment ↔ growth feedback)

We use the following minimal ODE skeleton (Blogger-ready; no MathJax):

(5.1a) $dA_k/dt = A_k \cdot (\lambda_k \cdot \cos D_k - \beta)$
(5.1b) $dD_k/dt = -\mu \cdot \sin D_k$

Interpretation:

λ_k is the mode's effective "fit advantage" under the protocol (data/task + current state).

β is a decay/cleanup pressure (weight decay-like, or any regularizing dissipative force).

μ sets the internal mismatch relaxation speed (how quickly the unit self-coheres).

Why this matches the modular-addition findings: the paper explicitly derives phase-mismatch dynamics that drive a doubled-phase difference toward 0 (alignment) and shows that alignment controls the effective growth behavior of learned Fourier components; together this creates a strong "self-reinforcing" selection dynamic.

5.2.1 Alignment improves growth

From (5.1a), growth is maximized when $D_k \approx 0$, since $\cos D_k \approx 1$. If $D_k \approx \pm\pi/2$, $\cos D_k \approx 0$ and growth stalls; if $D_k \approx \pi$, growth becomes negative unless λ_k is huge.

This captures the paper's empirical/mechanistic theme: modes with better internal phase relations become easier to amplify.

5.2.2 Growth accelerates alignment (via time allocation)

From (5.1b), D_k relaxes fastest when |sin D_k| is large. But crucially, in the actual learning system, "the mode that grows" increasingly dominates gradients/resources, so its mismatch variable is the one that most matters. This produces a **feedback loop**:

slightly better-aligned mode ⇒ slightly faster growth,

faster growth ⇒ mode dominates the unit's effective dynamics,

dominance ⇒ mismatch is corrected more decisively for that mode,

which further improves growth.

The modular-addition paper formalizes this behavior as part of its lottery-ticket analysis (competition between frequencies/components).

5.3 Winner-take-most collapse as a dynamical consequence

Define the "dominance ratio" of a candidate winner w relative to all others:

(5.2) $R(t) := A_w(t) / \Sigma_{j \neq w} A_j(t) \to +\infty$

This is the operational meaning of "collapse" in this paper: the system doesn't need other modes to vanish; it needs the winner to become **so dominant** that the rest are negligible under the observation map h and under downstream decision aggregation.

5.3.1 Why a single winner emerges (sketch)

Under mild separation assumptions (winner has a persistent advantage), define:

$\Delta\lambda := (\lambda_w \cdot \cos D_w^*) - \max_{j \neq w}(\lambda_j \cdot \cos D_j^*)$

where D_k^* denotes the effective mismatch after the fast relaxation transient.

Then for a wide set of trajectories, R(t) grows approximately exponentially:

$R(t) \approx R_{init} \cdot \exp(\Delta\lambda \cdot t)$

which yields the characteristic time-to-collapse estimate:

(5.3) $t_c \approx (1/\Delta\lambda) \cdot \log(R_{target} / R_{init})$

This matches the paper's central message about **lottery ticket selection**: small initial differences (amplitude or mismatch) can take time to amplify, but once the ratio crosses a threshold, dominance becomes rapid.

5.4 The "lottery ticket" content: what counts as a good ticket

Within this template, a mode k is a "good ticket" if it has one (or several) of:

higher initial amplitude $A_k(0)$,

smaller initial mismatch $|D_k(0)|$ (so $\cos D_k$ is larger sooner),

larger intrinsic advantage λ_k under the protocol.

The modular-addition paper emphasizes that **phase/mismatch** is not a minor detail: mismatch directly shapes whether a component grows or stalls, and training dynamics systematically reduce mismatch for the modes that are able to grow.

5.5 What is portable vs modular-specific

Portable (expected to recur in LLM-like systems)

Competition among candidate internal hypotheses (features/circuits/modes).

A coherence variable that gates growth (alignment, consistency, internal coupling).

Positive feedback producing delayed → sudden dominance.

A log-time dependence on initial advantage (5.3), explaining "why it takes many examples."

Modular-specific (do not over-literalize)

The exact meaning of "phase" as doubled-phase Fourier relations is task- and architecture-dependent.

The exact basis (Fourier) is privileged by group symmetry in modular arithmetic; other tasks have different natural bases.

5.6 Bridge to $\Xi(t) = (\rho, \gamma, \tau)$ (short preview)

This microdynamic template is the first place Ξ becomes mechanically grounded:

ρ increases when a small set of A_k concentrate into dominant modes.

γ increases when mismatch D_k collapses toward 0 (internal coherence rises).

τ is reflected in how quickly mismatch relaxes and how strongly non-winners are dissipated (μ, β-like effects), i.e., how sharply timescales separate between selection and cleanup.

We keep this as a preview; Section 6 will show how macro correctness can "snap" even when cross-unit alignment is not required (CWA), once enough collapsed micro-units contribute to an additive decision.

6. Macro Coherence Without Cross-Unit Alignment (CWA Layer)

This section introduces **Collapse Without Alignment (CWA)** as the macro-level complement

to Section 5's micro "collapse-by-competition." The key message is:

> **Even if micro units remain heterogeneous and mutually misaligned, a macro observable can become stable once it is an additive (or additivity-dominated) projection with sufficient symmetry / weak correlation.**

This is the precise sense in which a system can "suddenly understand" at the macro level: not because every micro part agrees, but because **the aggregate becomes high-SNR and therefore predictable.**

6.1 Precise definition: CWA = macro stability from additive projection

CAFT formalizes CWA as the widely observed regularity that **macro variables remain stable even when micro states are heterogeneous and misaligned**, provided the macro is an additive projection that **commutes with coarse-graining** (i.e., predictability survives aggregation without needing micro coordination).

We express the CWA backbone in the simplest "vote-sum" form:

(6.1) $Y := \Sigma_{i=1..M} v_i$

v_i can be any micro contribution ("votes," local estimators, partial features, small circuits).

Y is the macro observable used for decision or prediction (often after a final nonlinearity, e.g., argmax/softmax).

CWA claim (operational): under a declared protocol P, Y can be stable/predictive even when the v_i are not aligned to each other, as long as (i) the macro is additivity-dominated and (ii) the micro heterogeneity obeys cancellation-friendly structure (weak correlation or symmetry).

6.2 Variance accounting and the "why it becomes stable" mechanism

Start from exact variance decomposition:

(6.2) $\mathrm{Var}(Y) = \Sigma \mathrm{Var}(v_i) + 2\Sigma_{i<j} \mathrm{Cov}(v_i, v_j)$

This equation is the entire battleground:

If covariances are strongly positive and grow with M, the macro can become unstable (non-CWA regimes).

If covariances are small, cancel by symmetry, or are controlled, then Var(Y) grows slowly enough that signal dominates.

Define signal-to-noise ratio as:

(6.3) $\mathrm{SNR}(Y) \propto |E[Y]| / \sqrt{\mathrm{Var}(Y)}$

CWA scaling intuition (weak correlation / symmetry):

If $E[v_i] \approx \mu$ and $Var(v_i) \approx \sigma^2$ with $Cov(v_i,v_j) \approx 0$ (or symmetry-cancelled),
then $E[Y] \approx M\mu$ and $Var(Y) \approx M\sigma^2$, hence:

(6.4) $SNR(Y) \approx (M|\mu|) / \sqrt{(M\sigma^2)} = \sqrt{M}\cdot(|\mu|/\sigma)$

So macro predictability can jump sharply when $\sqrt{M}\cdot(|\mu|/\sigma)$ crosses a decision threshold—even if the individual v_i remain noisy, diverse, and mutually inconsistent in detail. This is the cleanest mathematical skeleton behind "collapse without alignment."

6.3 What "without alignment" means (and what it does NOT mean)

CWA does **not** mean "no structure." It means:

No requirement of cross-unit micro coordination (v_i do not need to agree on phases, features, or internal representations).

Yes requirement of cancellation-friendly macro conditions (additivity + weak correlation or symmetry in the ensemble).

In CAFT language: many macros are well-behaved because the projection suppresses micro chaos; CWA is the conservative regime where this suppression holds.

6.4 The modular-addition nuance: intra-unit alignment may be needed, cross-unit agreement is not

The modular-addition paper gives a rare, explicit demonstration of the nuance CWA needs:

6.4.1 Intra-unit alignment (often required)

Within a neuron, the paper reports and analyzes a **layer-wise phase coupling** ("doubled phase") and shows the phase mismatch is dynamically driven toward 0 (alignment as an attractor).

Interpretation: a micro unit may need internal coherence to become a reliable contributor v_i.

6.4.2 Cross-unit alignment (not required; diversity is essential)

Across neurons, the paper finds **model symmetry**: phases are approximately uniform within a frequency group and frequencies are covered across the population (diversification).

Mechanistically, each neuron contributes a **biased vote** whose residual terms depend on its own frequency-phase "view," but the network succeeds by **majority voting**: diversified, biased votes aggregate to cancel residual noise.

This is CWA in a concrete, fully worked example:

micro votes v_i are not mutually aligned in phase,

macro Y is stable because additive aggregation plus symmetry cancels noise.

6.5 CWA as the macro explanation for “sudden understanding”

Combine Sections 5 and 6:

Section 5 explains how each unit can **collapse onto a dominant mode** (making v_i more structured).

Section 6 explains why the system can still look “sudden” at the macro level: once enough structured-but-diverse v_i exist, the aggregate Y crosses an SNR threshold.

In one line:

(6.5) SuddenMacroStability occurs when SNR(Y(t)) crosses a protocol-dependent threshold Θ(P)

This is the macro layer of the story: **a sharp jump can be a statistical threshold event** even if micro-level changes are continuous.

Next (Section 7) we will package the forces that move the system across this threshold into the PORE operator channels **Pump–Probe–Switch–Couple**, so “sudden understanding” becomes a controllable, testable regime transition rather than an anecdote.

7. Operator Channels: Pump–Probe–Switch–Couple as Learning Control Grammar

This section turns “training forces” into a **small, testable control vocabulary**. Under PORE, operator channels are not metaphors; they are **declared intervention families** inside the protocol P = (B, Δ, h, u), and they must be **separable by signatures** in compiled coordinates $\hat{\Xi}$ and in the log stream z[n].

7.1 Operator channels as a protocol object (not an interpretation)

Under a fixed protocol, the admissible channel at each tick is:

(7.1) u[n] ∈ {Pump, Probe, Switch, Couple}

PORE’s insistence is operational:

you must declare what you changed (which channel, how much),

and later explanations must be reconstructible from the protocol log:

(7.2) z[n] = h(x(to + nΔ))

7.2 Why four operators are “enough” in many learning transitions

PORE’s four channels are introduced as a **minimal spanning set** of intervention families that target distinct generator mechanisms in an open-system view. Concretely, the framework states a canonical pairing at the generator level:

Pump targets gradient/potential deformation,

Probe targets current/circulation / interface coupling,

Switch targets a jump / regime-change channel,

Couple targets constraint penalty / binding closure.

A compact formal statement is:

(7.3) $\mathscr{L}_u = \mathscr{L}_0 + \Sigma_{i\in\{P,Q,Sw,C\}} u_i \mathscr{L}_i + \Sigma_{i<j} u_i u_j \mathscr{L}_{ij} + \dots$

Interpretation for learning systems:

Pump is any intervention that reshapes the effective “energy landscape” of fitting (drive, basin depth, resource injection).

Probe is any intervention that changes what is queried/measured/decoded, with explicit backreaction testing.

Switch is any discrete regime change (schedule step, optimizer swap, curriculum phase change, endogenous mode jump).

Couple is any intervention that increases closure/binding/coherence and reduces leakage (constraints, tying, consistency regularization, agreement enforcement).

This set is “enough” not because the world is simple, but because (7.3) explicitly allows **cross-talk terms** $\mathscr{L}_{ij}$; the four channels are a **first-order basis** for identifying what kind of change is happening.

7.3 Making it empirical: the local gain model in Ξ-space

To avoid narrative labeling (“it’s Pump-ish”), PORE provides a minimal experiment logic: estimate how each channel moves the compiled coordinates $\hat{\Xi}$.

Fix a protocol-valid window where $\hat{\Xi}$ is stable (Gate 1) and probing is identifiable (Gate 3). Then the local response model is:

(7.4) $\delta\Xi_{t+1} = \tilde{A}\,\delta\Xi_t + \hat{G}\,\delta u_t + \xi_t$

where:

$\delta\Xi_t := \hat{\Xi}_t - \bar{\Xi}$ is deviation from a local operating point $\bar{\Xi}$,

δu_t is the declared one-hot (or sparse) operator pulse in the 4-channel basis,

Ĝ is the **gain matrix** mapping operator pulses to coordinate shifts.

Column partition:

(7.5) $\hat{G} = [\hat{g}_P\ \hat{g}_Q\ \hat{g}_{Sw}\ \hat{g}_C] \in \mathbb{R}^{3\times 4}$

A practical dominance diagnostic is then:

(7.6) $i^*(t) := \arg\max_{i\in\{P,Q,Sw,C\}} \|\hat{g}_i\|_2$ (local operator dominance)

This is deliberately conservative: it treats "dominance" as an estimable effect direction, not a philosophical label.

7.4 The hard rule that keeps Probe honest (why CWA/"collapse" doesn't become measurement drift)

PORE requires that "Probe must not secretly be Pump/Switch/Couple." The harness makes this falsifiable:

If a null (small) Probe pulse materially moves Ξ̂ or changes drift, Gate 3 fails and Probe is not behaving as Probe.

The framework states the forced diagnosis:

(7.7) Gate3 fail ⇒ "Probe" is not identifiable as Probe; probing is acting like Pump/Switch/Couple

This matters directly for LLM "sudden understanding" discussions: without (7.7), it is too easy to confound "we measured differently" with "the system changed regime."

7.5 Operator Signature Table (deliverable)

The table below is written for **protocol logs**. "What to measure" refers to z[n] and compiled Ξ̂ (or proxies of it). "Dominates" means its signature is the clearest first-order driver in (7.4)–(7.6), not that other channels are absent.

Operator	What it *is* (control family)	Primary Ξ effect (expected signs)	What to measure (from z[n], Ξ̂)	Dominance patterns (tell-tale signs)

Operator	What it *is* (control family)	Primary Ξ effect (expected signs)	What to measure (from z[n], Ξ̂)	Dominance patterns (tell-tale signs)
Pump (P)	Fit-drive / landscape reshaping (resource injection, basin deepening)	typically ∂ρ/∂u_P > 0 (structure mass increases); may also increase τ if drive induces churn	training loss slope; norm/scale growth; representation concentration proxies (ρ̂-family); drift changes without discrete jumps	rapid training-loss decrease; smooth monotone growth of ρ̂; no discrete breakpoints; gains ĝ_P large in (7.5)
Probe (Q)	Measurement/ query changes (h changes, diagnostic readouts) with backreaction control	intended: small		ĝ_Q
Switch (Sw)	Regime change trigger (schedule step, optimizer swap, curriculum phase; endogenous mode jump)	Switch primarily changes τ through switching time / discrete event channel	breakpoint detection in curves; change-point in drift Ã; jump-rate proxies; hysteresis across runs	discontinuities / kinks; abrupt change in slope or variance; pre/post behavior not comparable without re-compilation; large ĝ_Sw or detected jump kernel change
Couple (C)	Closure/binding/ coherence enforcement (constraints, tying, agreement pressure)	∂γ/∂u_C > 0 and leakage decreases (Couple "locks in")	coherence proxies (γ̂-family); redundancy/ consensus across submodules; leakage/instability metrics; constraint violation rates	rising γ̂ with reduced volatility; decreased disagreement across components; improved CWA-style macro stability without requiring cross-unit alignment; large ĝ_C in (7.5)

Notes (so the table stays honest):

Channels are not perfectly orthogonal; cross-talk is expected and is explicitly modeled by higher-order terms in (7.3).

"Dominance" is always local-in-regime; Switch dominance often invalidates linear gain estimation across the breakpoint by design (you must segment regimes).

7.6 Why this operator grammar is the right bridge between micro-collapse and macro-CWA

Sections 5–6 gave two core mechanisms:

micro: winner-take-most mode selection (collapse-by-competition),

macro: additive cancellation yields stable outputs without cross-unit alignment (CWA).

The operator grammar is what makes these mechanisms **actionable**:

Pump accelerates or delays mode selection by changing drive strength and resource allocation.

Couple increases coherence/closure so micro-collapsed units become reliable contributors v_i.

Switch explains suddenness that is not purely emergent: the protocol itself can introduce jumps (schedules/curricula), and the model can also undergo endogenous mode switches.

Probe makes "understanding" testable while protecting against measurement-induced artifacts (Gate 3).

Next, Section 8 will use this grammar to formalize grokking as a **two-force competition** (Pump vs Switch/Couple-like cleanup) and connect it to staged trajectories in Ξ-space.

8. Grokking as a Two-Force Competition with Three Observable Phases

This section packages "grokking" (delayed → sudden generalization) as a **competition between two forces** acting on the compiled system, producing **three observable phases**. The modular-addition paper is unusually explicit: it designs a progress measure, runs weight decay, and then attributes the full timeline to **loss minimization vs weight decay**.

8.1 The control ratio κ(t): Drive vs Cleanup

We define the force ratio:

(8.1) κ(t) := Drive(t) / Cleanup(t)

Drive(t) is the effective "fit pressure" pushing the system to reduce training loss (Pump-like).
Cleanup(t) is the effective "structural refinement pressure" that prunes / regularizes / compresses (Switch/Couple-like).

In the modular-addition grokking setup, the paper identifies these two forces concretely as:

Drive ≈ loss minimization (cross-entropy gradient pressure),

Cleanup ≈ weight decay (explicit ℓ2 decay acting as a pruning/refinement force).

8.2 What makes the phases observable: progress measures

To track phase transitions, the paper monitors four progress measures during grokking experiments (train fraction 0.75, weight decay 2.0):

Train/Test loss & accuracy (macro regime indicator).

Phase difference via |sin(D⋆_m)| where D⋆_m := (2ϕ⋆_m − ψ⋆_m) mod 2π (coherence proxy).

Frequency sparsity via IPR of Fourier coefficients (structure concentration proxy).

ℓ2 norm of parameters (scale/complexity proxy; also indicates ongoing loss pressure vs decay pressure).

The paper then explicitly states: these measures reveal **three phases** driven by the two-force competition.

8.3 Phase I — Memorization (Drive-dominant)

Definition (operational): κ(t) ≫ 1, so Drive dominates; Cleanup is too weak to enforce a clean sparse structure.

Observed signatures in modular addition:

Training loss drops rapidly to ~0; training accuracy → 100%.

Parameter norms increase rapidly (Drive pushes scaling).

Internally, multiple frequency components can keep growing "in parallel" (a perturbed version of the lottery-ticket dynamics), yielding a **perturbed Fourier solution** that overfits.

The model exploits symmetry artifacts: it also performs well on test points whose symmetric counterparts are in training, but fails on truly unseen held-out points.

Intuition: in this phase, the system finds **a workable but dirty representation**: enough to drive training loss down, not yet "clean" enough to generalize.

8.4 Phase II — Transition / Generalization I (competition + rapid cleanup)

Definition (operational): κ(t) ≈ 1 in effect: Drive still acts, but Cleanup becomes strong enough to shape *which* structure survives.

The paper characterizes this phase as "precise interplay":

Norms continue to grow → Drive is still active.

But weight decay induces **frequency-domain sparsification**: it prunes non-feature components while the dominant component continues growing.

This pushes the representation closer to the **clean single-frequency solution**, and **test loss drops sharply**.

They also note a visible turning point in their run (around step ~10,000) separating this transition from the final stage.

Intuition: this is where "sudden understanding" *usually appears* at the macro level: not because

Drive suddenly became smarter, but because Cleanup finally becomes decisive enough to remove the residual noise terms and leave the generalizable structure.

8.5 Phase III — Refinement / Generalization II (Cleanup-dominant)

Definition (operational): $\kappa(t) \ll 1$: Cleanup dominates; Drive mainly maintains fit while refinement continues.

Observed signatures:

Weight decay becomes the dominant force.

Test accuracy improves more slowly toward perfection.

The system continues pruning/refining the remaining residual structure into the clean sparse Fourier representation.

Intuition: after the sharp transition, the remaining work is often "slow polishing": diminishing returns, but still structurally meaningful.

8.6 Connection to $\Xi(t) = (\rho, \gamma, \tau)$

We now connect the three phases to the intrinsic triple roles from Section 3. This is not claiming a unique estimator—only a consistent **role mapping** supported by the modular-addition progress measures.

8.6.1 $\rho(t)$ grows (structure mass concentrates)

In Phase I, ρ rises quickly as the system pours capacity into fitting (norm grows; frequency components amplify).

In Phase II–III, ρ becomes **more concentrated**: IPR increases as non-feature frequencies are pruned and single-frequency dominance strengthens.

8.6.2 $\gamma(t)$ reorganizes (coherence / lock-in increases)

The paper tracks coherence via $|\sin(D\star_m)|$ (layer-wise phase alignment).

Across phases, γ is best read as "how strongly the representation is locked into a consistent internal coupling pattern" (phase relations stabilize; dominant modes reinforce without rotation).

8.6.3 $\tau(t)$ separates (timescales split; cleanup becomes slower but decisive)

In Phase I, dynamics are fast: Drive rapidly reduces training loss while structure is still messy.

In Phase II, two timescales coexist: norms can keep growing (Drive) while frequency noise is pruned (Cleanup).

In Phase III, Cleanup dominates and becomes the slow governor: refinement continues over long time windows.

Operationally, τ is the coordinate that explains why you can get **long delays** followed by sharp transitions: timescale separation lets the system "sit" in a high-train/low-test basin until cleanup gradually clears a path to the generalizable basin.

8.7 One-line regime summary (usable later for the critical surface)

A minimal summary consistent with the paper's story is:

(8.2) Phase I: $\kappa(t) \gg 1 \Rightarrow$ memorize via perturbed structure
(8.3) Phase II: $\kappa(t) \approx 1 \Rightarrow$ rapid cleanup + sharp test improvement
(8.4) Phase III: $\kappa(t) \ll 1 \Rightarrow$ slow refinement toward clean general solution

This prepares Section 9, where we convert the above into a **critical-surface crossing condition** in Ξ-space (protocol-relative), i.e., the explicit mathematical form of "when the jump happens."

9. Critical Surface Model: When the Jump Happens

This section turns "sudden understanding" into a **protocol-relative crossing event** in Ξ-space. The core move is: define a **critical surface** $\Sigma_c(P)$ that separates qualitatively stable regimes (memorization-like vs generalization-like), and express the transition as a **minimal inequality** driven by the Drive–Cleanup competition and the micro→macro mechanisms established in Sections 5–8.

9.1 Critical surface $\Sigma_c(P)$ in Ξ-space (definition)

Recall $\Xi(t) := (\rho(t), \gamma(t), \tau(t))$ as compiled control coordinates.
Because Ξ is protocol-compiled, the regime boundary must also be protocol-relative:

(9.0) $\Sigma_c(P) \subset \mathbb{R}^3$ is the set of Ξ where the loop's macro behavior changes class under fixed P.

We now define a simple scalar "generalization control index":

(9.0a) $GCI(t;P) := \kappa(P)\cdot\rho(t)\cdot\gamma(t) / (\tau(t) + \varepsilon_0)$

where ε_0 is a small stabilizer to avoid division artifacts in low-τ regimes.

Then the **critical surface** is the level set:

(9.0b) $\Sigma_c(P) := \{ \Xi : GCI(\Xi;P) = \Theta(P) \}$

$\Theta(P)$ is a protocol-dependent threshold capturing task difficulty, evaluation criterion, and the chosen observation map h.

$\kappa(P)$ is the protocol-dependent effective drive/cleanup ratio (Section 8), abstracting "loss

minimization vs weight decay” in the modular-addition grokking setting.

9.2 Minimal sufficient inequality for regime switch

We write the regime condition exactly as the paper’s core equation requested:

(9.1) $\kappa(P)\cdot\rho(t)\cdot\gamma(t) / \tau(t) \geq \Theta(P) \Rightarrow$ generalization regime

Interpretation (tight, operational):

$\rho(t)$ must be high enough that **usable structure mass** exists (post-selection concentration).

$\gamma(t)$ must be high enough that structure is **coherently locked** (internal coupling is stable).

$\tau(t)$ must be low enough that the system’s structure is not continuously smeared by agitation/dephasing (timescale separation supports refinement).

$\kappa(P)$ weights how much the protocol is currently “pushing” vs “cleaning”—in modular addition grokking this is exactly the competition between loss minimization and weight decay that drives the three-stage timeline.

Why “minimal”? Because it only uses:

the three intrinsic coordinates (ρ, γ, τ) as order parameters, and

the single protocol-weighted ratio $\kappa(P)$,
and it does not assume a specific internal basis (Fourier is only the toy case).

9.3 Why it looks sudden: steep crossing induced by feedback loops

The inequality (9.1) can be crossed smoothly in Ξ-space while **observed performance jumps sharply**. There are three compounding reasons, each grounded in earlier sections and in the modular-addition laboratory.

9.3.1 Micro positive feedback makes ρ and γ accelerate near the boundary

Section 5’s “lottery ticket” mechanism is explicitly a competitive dynamic where the winning component grows exponentially faster once it has a small advantage (initial magnitude + smaller phase misalignment), producing winner-take-most dominance.
As the winner dominates:

ρ increases rapidly (mass concentrates into the dominant mode),

γ rises (mismatch collapses; internal coherence stabilizes).

So $d(\rho\cdot\gamma)/dt$ can spike even if the underlying parameter drift looks gradual.

9.3.2 Macro CWA thresholding makes the observable jump when SNR crosses a line

Section 6 showed that macro stability can “snap” when additive aggregation crosses an SNR threshold, even with cross-unit heterogeneity (CWA).
In modular addition, the network’s correctness emerges via population-level diversification + phase symmetry, enabling majority-voting cancellation of noisy neuron outputs—exactly a CWA-styled stabilization mechanism.
Thus, once enough micro units are “good voters,” the macro decision can flip from unstable to stable quickly.

9.3.3 τ-driven timescale separation explains long delay then sharp improvement

In grokking, the paper characterizes a three-stage process where weight decay eventually prunes residual noise and refines features into the sparse representation required for generalization.
That is precisely a τ-story: early on, the system can fit while still “hot”/messy; later, cleanup dominates and effectively lowers τ (or increases separation), allowing the crossing of Σ_c(P) to finally occur.

9.4 A convenient observable form (optional, for later empirical fitting)

To connect “surface crossing” to an observed metric like test accuracy, we can model generalization as a steep sigmoid of the control index:

(9.2) GenScore(t) ≈ σ(a·(GCI(t;P) – Θ(P)))

σ is a monotone squashing function (logistic-like).

a sets sharpness (large a ⇒ more sudden-looking).

This does **not** add a new assumption; it is just a standard way to represent that observed performance is often a thresholded readout of a latent order parameter.

9.5 Protocol discipline: Σ_c(P) is only meaningful if Ξ̂ is a legitimate chart

Because everything is protocol-relative, Σ_c(P) is only a valid object if the compiled coordinates pass minimal gates:

Gate 1: proxy stability (Ξ̂ behaves like a chart).

Gate 3: probe identifiability (measurement isn’t secretly changing the loop).

This is why the critical surface model is framed as Σ_c(P), not Σ_c: change P, and you are measuring a different object unless comparability is re-earned.

Next (Section 10) we will connect this “critical surface crossing” grammar to the SMFT-style projection/collapse pattern—kept operational—showing how “selection” can be treated as an internal operator in the loop rather than an external interpretation layer.

10. SMFT Bridge: Projection/Collapse as the Generic Grammar of Selection

This section provides a **minimal, operational** bridge: we treat "collapse" as the **generic grammar of selection** inside a protocol-closed loop, not as a metaphysical claim. The purpose is purely structural:

Section 5 gave **micro selection** (winner-take-most mode dominance).

Section 6 gave **macro stabilization** (CWA: additive projection yields stable macros without cross-unit alignment).

Section 9 packaged both as a **critical-surface crossing** in Ξ-space.

SMFT contributes one clean abstraction: **separate "field evolution" from "projection selection."**

10.1 Two-step grammar: evolution then projection (protocol-relative)

We define a latent "field state" Ψ(t) as *whatever high-dimensional object* represents the system's distributed potential at time t (for LLM training, think: a distribution over internal hypotheses/ circuits/features, not necessarily a literal physical field).

Protocol-conditioned evolution:

(10.1) $\Psi(t + \Delta) = U_P[\Psi(t)]$

U_P is the **protocol-relative update** induced by P = (B, Δ, h, u): optimizer dynamics, data exposure, regularizers, and any declared operator pulses. (This is the "pre-collapse propagation" role.)

Observer/proxy-conditioned projection:

(10.2) $\Psi \mapsto \Psi' := \Pi_o \Psi \,/\, ||\Pi_o \Psi||$

Π_o is a **projection operator** defined by the observer/proxy context o (what is being resolved/ selected at this tick: objective, evaluation, logging map, downstream decision rule). SMFT emphasizes that projection is an internal resolution step that selects a path/branch and writes a trace, rather than "measuring a hidden value."

Operational reading (no metaphysics):

(10.1) is "the system evolves under the training loop."

(10.2) is "the loop commits to a particular effective structure because only that structure survives the protocol's selection constraints."

10.2 Compatibility with Section 5 (mode competition as projection-like selection)

Section 5 modeled micro feature emergence as competing modes k with amplitudes A_k(t) and mismatch variables D_k(t), producing winner-take-most dominance. The modular-addition paper explicitly describes **frequencies competing within each neuron**, with the winner determined by initial magnitude and phase misalignment.

In the present grammar, that is simply:

U_P drives continuous competition and amplification (10.1).

Π_o is effectively "keep the component(s) that reduce loss under constraints," i.e., a selection map that makes one mode dominate the operational representation (10.2).

So "collapse" here does **not** mean an instantaneous discontinuity; it means that after repeated application of (10.1)+(10.2) over ticks, the representation becomes effectively supported on a small subset of modes (winner(s) dominate), matching the paper's neuron-wise "lottery ticket" mechanism.

10.3 Compatibility with Section 6 (CWA macro stabilization as projection + additivity)

CAFT's CWA states: macro observables can be stable even when micro states are heterogeneous/misaligned, provided the macro is an additive projection that retains predictability without micro coordination.

This matches the same grammar at a higher scale:

Let micro contributions be v_i (diverse, not aligned).

Let the macro constructor be an additive aggregator (a projection/observable in CAFT notation).

Then Π_o can be interpreted as the **macro readout projection**: it does not require micro alignment; it requires that the readout commutes with coarse-graining (CWA condition).

In the modular-addition paper, this is exactly the "majority-voting / noise cancellation" mechanism: single neurons are imperfect, but population symmetry/diversification makes the aggregate behave like an indicator after softmax.

10.4 Minimal bridge claim (what SMFT contributes here)

SMFT's "one assumption" framing (chaotic pre-collapse field is the only true hypothesis; wave-like form and observer-triggered resolution are emergent/constitutive) is used here only as a **discipline**: do not treat "selection" as an extra story; treat it as a necessary internal operator if you want a trace-realized world.

Concretely, SMFT supplies two operational ideas we reuse:

Projection is non-unitary and trace-making at the tick scale: it resolves alternatives into a

committed trajectory.

Observerhood can be treated as a class of projection operators with differing trace-awareness/recursion (useful later when we discuss probes, backreaction, and self-referential loops).

This integrates cleanly with PORE's Gate 3 rule ("probe must not secretly rewrite the loop"), because in both formalisms, a "measurement" that changes the future kernel is not a passive read—it is a projection intervention.

10.5 What we will (and will not) do with this bridge

We **will** use (10.1)–(10.2) as a compact way to describe *selection events* that correspond to Section 9's crossing of $\Sigma_c(P)$.

We **will not** claim that LLM training is quantum measurement, or that Ψ is a literal physical wavefunction. The only claim is that the **mathematical grammar "evolve then project"** is a useful universal template for understanding how "potential" becomes "committed structure" in protocol-closed learning loops.

Next (Section 11) we return to engineering: define **practical proxy families** for $\hat{\rho}, \hat{\gamma}, \hat{\tau}$ that can be computed from real LLM training logs (without requiring access to an explicit Fourier basis), while preserving the PORE gate discipline.

11. Practical Proxies: How to Estimate $\hat{\rho}, \hat{\gamma}, \hat{\tau}$ in Real LLM Training

This section provides **proxy families** for estimating the intrinsic triple $\Xi(t) := (\rho(t), \gamma(t), \tau(t))$ in real LLM training—where we usually *cannot* inspect a clean privileged basis (unlike modular addition's Fourier laboratory). The guiding rule is the one stated in the intrinsic triple framework:

> ρ, γ, τ are **role-defined** and only require **monotonicity** with respect to intended role; estimators are domain- and protocol-dependent.

We therefore give multiple **proxy families** and a **plug-in checklist**. There is **no privileged choice**: you must declare your proxy set as part of the protocol $P = (B, \Delta, h, u)$ and validate it via Gate 1 (proxy stability) and Gate 3 (probe backreaction).

11.1 Proxy discipline (PORE rules)

11.1.1 Windowed compilation (required)

Compile proxies on sliding windows W_k:

(11.1) $\hat{\Xi}(W_k) := (\hat{\rho}(W_k), \hat{\gamma}(W_k), \hat{\tau}(W_k))$

11.1.2 Gate 1: stability (required)

Use coefficient-of-variation stability over windows:

(11.2) $CV_x := \mathrm{std}(\{x(W_k)\})/(|\mathrm{mean}(\{x(W_k)\})|+\varepsilon_0)$, $x \in \{\hat{\rho},\hat{\gamma},\hat{\tau}\}$
(11.3) Gate1 pass $\Leftrightarrow CV_\rho \le c_\rho \wedge CV_\gamma \le c_\gamma \wedge CV_\tau \le c_\tau$

Failing Gate 1 forces a protocol action (revise B/Δ/h or segment regimes); it does *not* permit narrative patching.

11.1.3 Gate 3: probe backreaction sanity (required)

A "Probe" must not silently act like Pump/Switch/Couple. Run a declared null probe δu_Q and require small coordinate change:

(11.4) $\Delta\hat{\Xi}_Q := \hat{\Xi}_Q - \hat{\Xi}_0$
(11.5) Gate3 pass $\Rightarrow ||\Delta\hat{\Xi}_Q||_2 \le \theta_Q$ (and jump-rate not increased)

11.2 $\hat{\rho}$ proxy families (representational mass / concentration)

Role: ρ increases when "usable structure mass" is **concentrated/occupied**, not merely when weights are large.

Family ρ-A: spectral concentration (needs checkpoints)

Pick one or more weight matrices W_ℓ (or attention/output projections). Let singular values be $s_1 \ge \ldots \ge s_r$.

(11.6) $\hat{\rho}_{\mathrm{spec},k}(W_\ell) := (\Sigma_{i=1..k} s_i^2)/(\Sigma_{i=1..r} s_i^2)$

Interpretation: higher $\hat{\rho}_{\mathrm{spec},k}$ means more energy concentrated into top directions (more "structure mass" in fewer modes).

Optional "effective rank" (smooth concentration proxy):

(11.7) $p_i := s_i^2/(\Sigma_j s_j^2)$, $r_{\mathrm{eff}} := \exp(-\Sigma_i p_i \log p_i)$
(11.8) $\hat{\rho}_{\mathrm{erank}} := 1/(r_{\mathrm{eff}}+\varepsilon_0)$

Family ρ-B: activation concentration (no weights, needs hidden-state logs)

For a layer's hidden states H (token×dim), compute covariance eigenvalues λ_i:

(11.9) $\hat{\rho}_{\mathrm{act},k} := (\Sigma_{i=1..k} \lambda_i)/(\Sigma_{i} \lambda_i)$

Interpretation: as representations "crystallize," activity often concentrates into fewer stable directions.

Family ρ-C: compression / description-length proxies (loss-only friendly)

When you cannot access weights/activations, use “compression-like” proxies that track whether the learned solution becomes **simpler** while preserving fit—this is the operational signature of grokking’s refinement stage.

Examples (choose one):

normalized minimum validation loss under a fixed regularization budget,

MDL-like proxies using code-length of a probe model trained on frozen representations,

parameter norm growth *minus* effective-rank drift (if both are available) to avoid conflating scale with structure.

(11.10) $\hat{\rho}_{DL} := -\widehat{DL}(W_k)$ (monotone: lower description length ⇒ higher $\hat{\rho}$)

11.3 $\hat{\gamma}$ proxy families (coupling / coherence / lock-in)

Role: γ increases when the system becomes more **symmetry-locked / coherent**—i.e., subsystems reinforce a shared basis and resist diffusion. γ is explicitly not unique and must be role-monotone.

Family γ-A: cross-layer representational alignment (CKA / cosine stability)

If you can log hidden states, compute representational similarity between adjacent layers or between times.

(11.11) $\hat{\gamma}_{CKA} := \mathrm{mean}_{\ell \in L}\, \mathrm{CKA}(H^{\ell}(t), H^{\ell}(t-\Delta))$

Interpretation: rising $\hat{\gamma}_{CKA}$ indicates “lock-in” (less representational churn), consistent with stronger coupling.

Family γ-B: multi-head / multi-path agreement (redundancy as coherence)

Train small decoders (probes) from multiple locations (layers/heads) to predict the same target (next-token, task label, or a held-out diagnostic). Let predictions be $\hat{y}_a$, $\hat{y}_b$.

(11.12) $\hat{\gamma}_{red} := \mathrm{mean}_{a<b}\, \mathrm{corr}(\hat{y}_a, \hat{y}_b)$

Interpretation: higher redundancy means multiple subsystems carry consistent information—macro robustness can grow without forcing micro alignment (CWA-compatible).

Family γ-C: constraint/closure strength proxies (regularizers and consistency penalties)

If your protocol includes explicit coupling terms (e.g., consistency regularization, KL penalties, weight tying strength), use the measured penalty response as a γ proxy:

(11.13) $\hat{\gamma}_{cons} := \mathrm{mean}_W\, (\mathrm{Penalty_value} / (\mathrm{Penalty_ref} + \varepsilon_0))$

This aligns with γ’s original role as “domain-lock / boundary strength” aggregating confinement

mechanisms.

11.4 $\hat{\tau}$ proxy families (agitation / dephasing / timescale separation)

Role: τ increases when agitation/noise/churn smears structure and destroys coherence; τ is the "dephasing axis."

Family τ-A: grokking delay / two-timescale separation (loss curves)

Define two times: t_{fit} when training loss crosses a fit threshold, and t_{gen} when generalization crosses a generalization threshold.

(11.14) $\hat{\tau}_{delay} := t_{gen} - t_{fit}$

Interpretation: a large $\hat{\tau}_{delay}$ indicates strong timescale separation—classic grokking signature (fast fit, slow generalization refinement), matching the modular-addition three-stage story.

Family τ-B: churn / volatility in parameters or representations (checkpoints or activations)

If you log checkpoints:

(11.15) $\hat{\tau}_{churn} := \text{mean}_W\left(||\Delta W||_F / (||W||_F + \varepsilon_0)\right)$

If you log activations:

(11.16) $\hat{\tau}_{drift} := \text{mean}_W\left(1 - \text{CKA}(H(t), H(t-\Delta))\right)$

Higher churn/drift ⇒ higher $\hat{\tau}$.

Family τ-C: "recover vs switch" + jump/KL gates (PORE-native)

The intrinsic triple framework explicitly recommends $\hat{\tau}$ as "time to recover vs time to switch regimes," using jump/KL detection.

Let $\hat{r}_J(W)$ be jump-rate in window W (your chosen diagnostic), and $\hat{T}_{rec}$ be recovery time after a small pulse (MEP-style). Then:

(11.17) $\hat{\tau}(W) := \max\left(\hat{T}_{rec}(W), 1/(\hat{r}_J(W) + \varepsilon_0)\right)$

This is the most PORE-consistent τ proxy when you can run controlled pulses and need explicit regime-change hygiene.

11.5 "No privileged choice": how to choose a proxy set (plug-in checklist)

Below is a **plug-in checklist**. Pick the first row that matches what you can log. Then declare it as part of protocol P and validate Gate 1 and Gate 3.

Proxy Set A — Minimal logging (loss curves + schedule only)

You can log: train loss, eval loss/acc, learning rate, weight decay, step/epoch.
Compute:

$\hat{\rho}_A$:= −EvalLoss (or +EvalAcc) after normalizing by baseline

$\hat{\gamma}_A$:= redundancy of outcomes across seeds or across small probe heads (if available)

$\hat{\tau}_A := \hat{\tau}_{delay}$ from (11.14)

Use when: production training where only scalar metrics are kept.

Proxy Set B — Checkpoint-aware (weights available)

You can log: periodic checkpoints for selected layers.
Compute:

$\hat{\rho}_B$:= mean $\hat{\rho}_{spec,k}(W_\ell)$ from (11.6) or $\hat{\rho}_{erank}$ from (11.8)

$\hat{\gamma}_B$:= stability of top subspaces across time (CKA on projected activations, or cosine similarity of top singular vectors)

$\hat{\tau}_B := \hat{\tau}_{churn}$ from (11.15) + $\hat{\tau}_{delay}$ from (11.14)

Use when: you want "structure mass" proxies without logging activations.

Proxy Set C — Representation-aware (activations/hidden states available)

You can log: hidden states from selected layers on a fixed diagnostic batch.
Compute:

$\hat{\rho}_C$:= activation concentration $\hat{\rho}_{act,k}$ from (11.9)

$\hat{\gamma}_C := \hat{\gamma}_{CKA}$ from (11.11) + $\hat{\gamma}_{red}$ from (11.12)

$\hat{\tau}_C := \hat{\tau}_{drift}$ from (11.16) + $\hat{\tau}_{delay}$ from (11.14)

Use when: you want the cleanest "coherence/lock-in" signals.

Proxy Set D — Operator-lab (pulses + gate hygiene)

You can do: minimal experiment protocol with tiny one-channel pulses and jump/KL rejection.
Compute:

$\hat{\rho}_D$, $\hat{\gamma}_D$, $\hat{\tau}_D$:= as above, but **validate** each under Gate 1 and Gate 3

$\hat{\tau}_D := \hat{\tau}(W)$ from (11.17)

Optionally estimate gains $\hat{G}$ locally to identify which operator channel dominates regime motion.

Use when: you want causal, disentangled "why did it jump" diagnostics.

11.6 A caution that matters for LLMs (CWA compliance is a test, not an assumption)

Because Section 6 relies on a CWA-style additive stabilization story, you should treat "additivity-dominant macro" as something to **diagnose**, not assume. CAFT explicitly proposes empirical diagnostics (including permutation tests) to check whether a macro behaves like a CWA projection under coarse-graining.

We will turn this into concrete harness checks in Section 12.

Next (Section 12) we formalize the falsifiability harness: the minimum gates that prevent the integrated story from turning into "everything explains everything," especially when probing and regime switches can be confounded.

12. Falsifiability Harness: Gates That Prevent Self-Deception

A theory that can "explain everything" is usually a story generator unless it ships with **hard rejection tests** and **forced failure routing**. The Minimal Intrinsic Triple / Ξ-stack explicitly frames its harness as "falsifiable diagnostics, not metaphor," and enumerates standard gates plus a failure router that outputs concrete corrective actions.

In this paper, we adopt that harness and align it to our learning setting (LLM sudden understanding), with the exact gate numbering requested:

Gate 0: loop existence + boundary sanity

Gate 1: proxy stability

Gate 2: probe backreaction

Gate 3: control effectiveness

(Reference mapping: in the intrinsic triple harness, "boundary accounting" is a dedicated gate and "control effectiveness" is the final gate; we fold boundary accounting into Gate 0 as the boundary sanity requirement, and keep the backreaction/effectiveness tests intact.)

12.0 Harness inputs (minimum log you must be able to produce)

For any window W, you must be able to compute:

$\hat{\Xi}(W) = (\hat{\rho}(W), \hat{\gamma}(W), \hat{\tau}(W))$

jump / KL-like regime indicators (to reject windows contaminated by switches)

operator logs u(t) with channel IDs {Pump, Probe, Switch, Couple}

at least one probe signature observable (to test "Probe is not control")

If you cannot produce these, you can still train models—but you cannot claim you have an **engineering-grade explanation** for "sudden understanding" under a fixed protocol.

12.1 Gate 0 — Loop existence + boundary sanity

Purpose

Reject cases where "the system" is ill-defined: the boundary B is wrong, the loop is not recurrent/stable enough to admit compiled coordinates, or boundary fluxes dominate what you are calling internal learning.

Gate 0A: boundary accounting sanity (open-system check)

The intrinsic triple harness requires you to pay a **flux tax**: if the load-like coordinate ρ is being driven by boundary inflows/outflows you didn't model, any internal story is suspect.

(12.1) $\dot{\rho} = \Phi_{in} - \Phi_{out} + \dot{\rho}_{internal} + residual$
(12.2) $residual := \dot{\rho} - (\Phi_{in} - \Phi_{out} + \dot{\rho}_{internal})$
(12.3) $Gate0\ pass \Rightarrow E[|residual|] \le \varepsilon_B$

What failure looks like

Your "$\hat{\rho}$ increased" claim disappears when you account for batch size, curriculum shift, data repeats, or logging changes.

Sudden jumps track a hidden boundary change (e.g., data mixture swap, eval set swap, optimizer state reset).

How to revise protocol

Shrink or redraw boundary B (explicitly include the missing flux channel).

Redefine $\hat{\rho}$ so it aligns with a budgeted quantity under the boundary.

Gate 0B: loop validity (minimal recurrence/stability)

The Ξ-stack routine assumes you are analyzing a "loop segment" where a stable return-map-like behavior and bounded leakage are meaningful (otherwise regime labels are arbitrary). The harness lists loop validity metrics as required inputs.

What failure looks like

"Sudden understanding" coincides with unstable dynamics (exploding/vanishing metrics), or with nonstationary logging conditions.

How to revise protocol

Split the run into separate loop segments; do not fit a single Ξ-trajectory across mixed regimes.

12.2 Gate 1 — Proxy stability under minor perturbations

Purpose

Reject cases where $\hat{\Xi}$ is a moving target because proxies are not stable under the protocol. The intrinsic triple harness states proxy stability is the first gate: if it fails, “derived dynamics, gains, or regime labels are not meaningful.”

Gate definition (repeatability / drift)

(12.4) Gate1 pass ⇒ $\mathrm{Var}(\hat{\rho} \mid \text{protocol}) \le \varepsilon_\rho$, $\mathrm{Var}(\hat{\gamma} \mid \text{protocol}) \le \varepsilon_\gamma$, $\mathrm{Var}(\hat{\tau} \mid \text{protocol}) \le \varepsilon_\tau$

Protocol repeatability form:
(12.5) $\{\hat{\Xi}^{(k)}\}_{k=1..n}$:= repeated extraction under the same π
(12.6) Gate1 pass ⇒ $\mathrm{Var}_k(\hat{\rho}^{(k)}) \le \varepsilon_\rho \wedge \mathrm{Var}_k(\hat{\gamma}^{(k)}) \le \varepsilon_\gamma \wedge \mathrm{Var}_k(\hat{\tau}^{(k)}) \le \varepsilon_\tau$

What failure looks like

Small changes in random seed, windowing, diagnostic batch, or logging cadence flip the inferred regime.

Your “critical surface crossing” disappears when you rerun the same protocol.

How to revise protocol

Change proxy family ($\hat{\rho}/\hat{\gamma}/\hat{\tau}$) or refine h (observation map).

Change Δ (timebase) or split into sub-regimes where proxies are locally stable.

12.3 Gate 2 — Probe backreaction detection

Purpose

Reject cases where “measurement” is secretly acting as control. The intrinsic triple harness treats probing as an operator channel and requires a probe-on/probe-off test; if probing changes dynamics materially, you must upgrade the effective law to include an observer-coupling term.

Gate definition (probe-on/probe-off)

Fix boundary B and control u(t), change only probing intensity $\kappa(t)$:

(12.7a) $\Delta\Xi_{on} := \Xi(t_1) - \Xi(t_0)$ under $\kappa = \kappa_{on}$
(12.7b) $\Delta\Xi_{off} := \Xi(t_1) - \Xi(t_0)$ under $\kappa = \kappa_{off}$
(12.7c) Gate2 fail if $\|\Delta\Xi_{on} - \Delta\Xi_{off}\| \ge \theta_{\hat{O}}$

Required model update if Gate 2 fails
(12.8) $\dot{\Xi} = \ldots + C_{\hat{O}}(\Xi,t;\kappa(t)) + \ldots$

What failure looks like (LLM setting examples)

Evaluation choices change training trajectory (reward hacking, data filtering by eval, implicit early stopping heuristics).

“Understanding” appears only under a specific probing pipeline, not under probe-reduction controls.

How to revise protocol

Declare probing as an explicit coupling term (model the backreaction), or

redesign probes to reduce backreaction until Gate 2 passes (e.g., frozen evaluation batches, reduced adaptive probing).

12.4 Gate 3 — Control effectiveness (operator toggles must move Ξ as predicted)

Purpose

This paper is not only explanatory; it is operational. If you claim “this operator causes the jump,” you must demonstrate that toggling operators moves Ξ in the predicted direction and reduces deviation from the target regime. The intrinsic triple harness defines a minimal expectation test for effectiveness and a structured failure ladder.

Gate definition (before/after effectiveness)

Let $\delta\Xi := \Xi - \Xi^*$ for a desired target Ξ^* (or acceptable region). Under a candidate controller $u = -K\delta\Xi$:

(12.9a) $E_{pre} := E[\|\delta\Xi\| \mid t \in I_{pre}]$
(12.9b) $E_{post} := E[\|\delta\Xi\| \mid t \in I_{post}]$
(12.9c) Gate3 pass $\Rightarrow E_{post} \le E_{pre} - \Delta_{min}$

Minimal operator causality: local gain test (MEP)

When feasible, estimate a local gain map in Ξ-space using one-channel pulses:

(12.10) $\delta\Xi_{t+1} = \tilde{A}\,\delta\Xi_t + \hat{G}\,\delta u_t + \xi_t$
(12.11) $\Delta\Xi_t \approx \hat{G}\,\delta u_t$ (difference-out drift when valid)

Reject jump-contaminated samples (do not fit gains across switches):

(12.12) reject if $\|\Xi_{t+1} - \Xi_t\| \ge \theta_{KL}$

What failure looks like

Turning up “cleanup” (decay / compression pressure) does not shift $\hat{\tau}$ or improve generalization.

Increasing “couple” constraints does not increase $\hat{\gamma}$ or reduce leakage/disagreement.

Reported operator dominance flips across minor protocol-preserving reruns.

How to revise protocol (forced failure ladder)
The harness forces a structured diagnosis rather than ad hoc excuses, including: regime mismatch (unmodeled switching), proxy drift, boundary misaccounting, backreaction ignored, gain estimation unreliable, or Σ-level model class too weak.

12.5 Failure router (what you must do when a gate fails)

The intrinsic triple harness supplies a standard "route" output—explicit corrective actions rather than narrative repairs:

(12.13) Route ∈ {shrink boundary, change proxy, split loop, change timescale, elevate model class}

In this paper's context, interpret routes as:

shrink boundary: explicitly include missing exogenous channels (data mixture, eval policy, scheduling).

change proxy: switch to a different $\hat{\rho}/\hat{\gamma}/\hat{\tau}$ family that is role-monotone and stable.

split loop: segment phases (memorization/transition/refinement) instead of fitting one law across all.

change timescale: adjust Δ or window length W; recompile $\hat{\Xi}$ accordingly.

elevate model class: if local linear gain models fail repeatedly, add a Switch/jump kernel model or explicitly include backreaction term C_Ô.

Next (Section 13) we use this harness to state **predictions and interventions**: how Pump/Couple/Switch changes should shift κ(t), move Ξ(t), and therefore advance or delay the crossing of Σ_c(P)—with explicit "what would refute this" clauses.

13. Predictions and Testable Interventions

This section translates the integrated model into **testable predictions**. Each prediction is stated as:

a **protocol action** (what operator channel you toggle),

a **Ξ-space expectation** (how $\hat{\rho}$, $\hat{\gamma}$, $\hat{\tau}$ should move),

an **observable consequence** (what changes in generalization curves),

and **refutation patterns** (what would falsify the claim under the harness).

The modular-addition paper is useful here because it gives a concrete grokking setting where the competition between loss minimization and weight decay is explicit, and where diversification/phase symmetry explains population-level correctness.

13.1 Prediction class A — Increase cleanup β ⇒ earlier / cleaner generalization

Intervention (Switch/Pump mix)

Increase β (explicit weight decay; or any compression/refinement pressure) while holding boundary B and evaluation h fixed.

Model expectations

κ(t) = Drive/Cleanup decreases (Section 8).

τ̂ decreases (less agitation or stronger timescale separation enabling refinement).

ρ̂ concentration increases earlier (pruning accelerates dominance of reusable structure).

Observable consequence

Earlier onset of test improvement (t_gen decreases; grokking delay shrinks).

Cleaner transition: fewer oscillations/false starts; less "perturbed solution" residue.

In modular addition language: earlier move into the sparse Fourier regime (higher IPR earlier; stronger phase-lock indicators).

Refutation patterns (falsify under Gate 0–3)

Any of the following refutes the prediction under a stable protocol:

Raising β does **not** reduce τ̂_delay or does not move the breakpoints earlier.

Raising β increases training instability without improving or advancing generalization, *and* Gate 0 boundary sanity does not attribute this to boundary leakage.

The effect is non-monotone across minor perturbations (Gate 1 fails), implying the proxy or segmentation is invalid.

13.2 Prediction class B — Increase width/diversification ⇒ stronger CWA cancellation

Intervention (Pump/Couple via capacity + ensemble diversity)

Increase width M (or the effective number of parallel contributors, e.g., heads, experts) while holding the protocol otherwise fixed. Encourage diversification explicitly if possible (e.g., mild symmetry-breaking noise, anti-correlation penalties, or initialization variety—still inside declared P).

Model expectations

Cross-unit alignment is **not required**; what matters is that micro contributors provide sufficiently independent/structured votes v_i.

Var(Y) is controlled by covariance terms (Section 6), and weak correlation / symmetry implies SNR grows like √M.

Therefore the macro stability threshold in (9.1) is crossed earlier or more reliably.

Observable consequence

Higher final generalization reliability at fixed training budget.

Reduced variance across seeds.

Earlier “snap” once enough structured units exist (especially for tasks that can be represented by additive ensemble-like readouts).

In modular addition, this corresponds to the paper’s story: diversification across frequencies and phase symmetry enable majority-voting cancellation; wider models approximate those conditions better.

Refutation patterns

Increasing width yields no improvement in generalization timing or stability, *and* covariance diagnostics show no reduction (CWA condition doesn’t hold).

Width increases but macro stability worsens systematically—suggesting strong positive covariance between contributors (Var(Y) dominated by covariance terms). This refutes the CWA layer as the explanation for this setting.

13.3 Prediction class C — Modify curriculum ⇒ change κ(t) trajectory and mode-collapse time t_c

Intervention (Switch)

Change the curriculum: reorder examples, adjust difficulty schedule, or alter the mix of pattern families over time—while keeping evaluation and boundary consistent.

Model expectations

Curriculum changes affect:

the effective λ_k landscape of mode competition (Section 5),

the Drive/Cleanup ratio κ(t) trajectory (Section 8),

and thus the time-to-dominance t_c in the collapse model:

(13.1) $t_c \approx (1/\Delta\lambda)\cdot\log(R_{target} / R_{init})$

If early curriculum increases $\Delta\lambda$ for the "good" mode or increases initial advantage R_{init} (by repeatedly reinforcing a coherent basis), then t_c shortens; if it introduces competing hypotheses with similar $\Delta\lambda$ and strong interference, t_c lengthens.

Observable consequence

Earlier or later sudden transition depending on whether curriculum makes the intended basis easier to lock.

Potentially sharper transitions if curriculum reduces mode interference and helps coherence γ rise faster.

Refutation patterns

Curriculum changes have no measurable effect on $\kappa(t)$, $\hat{\tau}_{delay}$, or inferred mode-selection indicators, across stable repeats (Gate 1 pass).

Observed differences are fully explained by boundary leakage (e.g., evaluation set shifted or data duplication changed) (Gate 0 fail), meaning the intervention was not clean.

13.4 Stronger, falsifiable "operator signatures" (cross-check predictions)

Using the operator signature table (Section 7), the model predicts:

If **cleanup** is the driver of grokking, then increasing cleanup should move $\hat{\tau}$ and should move the location of regime switch; if not, your attribution is wrong.

If **CWA cancellation** is the macro mechanism, then increasing the number of contributors should increase SNR-like proxies (lower variance across runs, improved stability), not just lower training loss.

If **mode competition collapse** is the micro mechanism, then changes that affect $\Delta\lambda$ or R_{init} should predictably shift t_c-like delay measures.

These are not "nice-to-have"; they are the causal tests that keep the integrated story from being narrative-only.

13.5 Hard refutations of the integrated model (what would force revision)

The following patterns—if observed under a protocol that passes Gate 0–3—force revision of the model (not just parameter retuning):

No critical surface behavior: generalization improvements occur smoothly with no threshold-like transition in any compiled proxy space, across many tasks and settings.

No mode-selection signature: there is no evidence of winner-take-most concentration ($\hat{\rho}$ never concentrates) even when grokking-like curves appear.

Anti-CWA behavior dominates: macro variance increases with width (covariance terms dominate), producing worse stability as M grows. This breaks the CWA layer.

Probe confounds: changing measurement h changes the training trajectory (Gate 2 failure) but the analysis ignores the backreaction term—invalidating causal claims.

Operator toggles don't work: controlled Pump/Switch/Couple toggles do not move $\hat{\Xi}$ in predicted directions (Gate 3 failure), implying the operator grammar is not capturing the system's effective control coordinates.

Next (Section 14) we summarize what the integration explains that each component alone does not, and we list limitations and expected failure modes (where the basis is not clean, tasks are multi-modal, or interference breaks CWA assumptions).

14. Discussion: What the Integration Explains That Each Part Alone Doesn't

This section states what the integrated frame explains **that no single component explains on its own**, and where it should predictably fail.

14.1 Why "sudden" is natural (critical surface crossing)

Modular-addition dynamics alone can show "grokking happens" and that Fourier features and phase relations emerge.
But without the **critical surface model** (Section 9) and the **macro CWA layer** (Section 6), it is easy to misread suddenness as "a new capability appeared."

The integration gives a generic mechanism:

Micro: winner-take-most mode selection yields nonlinear acceleration (ratio R(t) grows slowly, then rapidly).

Macro: additive aggregation becomes stable when SNR crosses a threshold, creating a sharp observable jump even if micro change is continuous.

$\Xi + \Sigma_c(P)$: these combine into a **crossing event**: $\kappa(P)\cdot\rho\cdot\gamma/\tau \geq \Theta(P)$.

So "sudden" is not mysterious; it is the generic shape of a thresholded observable under positive feedback + cancellation.

14.2 Why "many examples" matters (time for mode selection + macro SNR)

A common confusion is: "If the model has capacity, why doesn't it generalize immediately?"

The integrated answer has two independent time requirements:

14.2.1 Micro time: selection needs time to amplify small initial advantages

Section 5's collapse-by-competition template implies:

dominance time t_c scales like log(R_target/R_init)/Δλ,

so even a correct mode can take long to dominate if Δλ is small or if initial advantage is tiny.

This naturally produces long "plateaus" where the model fits but hasn't consolidated into a generalizable basis.

14.2.2 Macro time: enough structured contributors must accumulate for CWA cancellation

Even if micro units are individually improving, the macro decision may not flip until:

enough micro contributors v_i are "good enough," and/or

covariance terms shrink enough that SNR grows.

Thus the model can be internally "on its way" while externally still failing.

In modular addition, the paper's population-level diversification + phase symmetry formalizes the condition under which many imperfect neurons collectively implement the rule.

14.3 Why interpretability can be made portable (PORE compilation)

Modular addition is unusually interpretable because symmetry picks Fourier as a natural basis. That can mislead people into thinking interpretability is "toy-only."

The integration adds a portability mechanism:

PORE says you don't need the "true basis"; you need a **protocol-defined observation map h** and **stable compiled coordinates** ($\hat{\Xi}$) with gates.

CWA says you don't need micro alignment; you can diagnose macro stability via aggregation tests and covariance signatures.

The critical surface model says "understanding" is a **regime boundary event**, so interpretability is about **finding the right order parameters**, not reading individual weights.

In short: modular addition supplies a calibration example of what good compiled coordinates look like; PORE supplies the general method for finding and validating analogues in real LLM training.

14.4 What the integration adds beyond "grokking explanations" alone

Many grokking accounts say: "weight decay makes solutions simpler." True, but incomplete.

The integrated frame adds:

A **control grammar** (Pump–Probe–Switch–Couple) that turns explanation into intervention.

A **macro stability principle** (CWA) that explains why the final behavior can be correct without internal uniformity.

A **hard harness** (Gate 0–3) that prevents interpretability narratives from surviving when probes backreact or proxies drift.

Together, these make the explanation operational and falsifiable rather than retrospective.

14.5 Limitations and expected failure modes

14.5.1 When symmetry/basis is not clean

Modular addition works because there is a privileged basis (Fourier) with strong invariance structure. In tasks without clean symmetries:

"mode" may not be separable,

multiple partially-correct bases may coexist,

selection may not be winner-take-most.

Prediction: the critical surface still exists, but proxies for $\rho/\gamma/\tau$ become noisier and Gate 1 stability is harder to satisfy.

14.5.2 Deep nets: multi-layer interactions and nonlocal coupling

The toy dynamics emphasize neuron-wise selection and simple phase relations. In deep transformers:

selection may happen at circuit level, not neuron level,

coherence γ may involve cross-layer routing and attention-mediated coupling,

τ may be dominated by interference between tasks/features.

Prediction: the operator grammar still helps, but the micro collapse model may require "multi-mode bundles" rather than single-mode winners.

14.5.3 Multi-task interference breaks CWA conditions

CWA depends on weak correlation / symmetry cancellation. In multi-task LLM training:

contributors may become strongly correlated (shared failure modes),

covariance terms in Var(Y) can dominate,

adding width can increase correlated noise rather than cancel it.

Prediction: in these regimes, “more capacity” may not improve stability; you need explicit Couple/Switch interventions to decorrelate contributors.

14.5.4 Probe backreaction is a practical hazard

Real training pipelines often use evaluation for early stopping, filtering, RLHF reward shaping, curriculum gating, etc. That makes “Probe” part of the control loop. If Gate 2 fails and you ignore it, your interpretability story becomes unreliable by definition.

14.6 Bottom line

The integration yields a compact, testable worldview:

Sudden understanding is a **regime transition** in compiled order parameters, not magic.

Long delays arise because micro selection and macro stabilization have independent timescales.

Interpretability is portable when treated as **protocol compilation + gates**, not as a hunt for a universal hidden basis.

The framework predicts where it will fail: messy invariances, deep coupling, multi-task covariance, and probe backreaction.

If you want, next I can draft the **Appendix A “Protocol Card Template”** and **Appendix B “Ξ-Proxy Cookbook”** in copy/paste form, so you can immediately apply the paper’s method to real LLM training runs.

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Appendix A. Protocol Card Template (copy/paste)

This appendix provides a **portable Protocol Card** in the PORE style. It is meant to be copied into experiment logs, OSF pages, lab notebooks, or code repos. The goal is to make “sudden understanding” claims **protocol-grounded and reproducible**.

A1) Protocol Identifier

Protocol ID: P-_____
Version: v_____
Date: _____
Owner: _____
Notes: (What changed since last version? Why?)

A2) Protocol Definition

(A.1) P := (B, Δ, h, u)

B — Boundary (what is inside the loop)

Inside (included):

Model architecture: (Transformer / MLP / etc.), depth, width, vocab, parameter count.

Optimizer + state: (AdamW / SGD / etc.), β params, momentum buffers, schedules.

Training data exposure rules: sampling, shuffling, curriculum, augmentation.

Regularizers: weight decay, dropout, label smoothing, KL penalties, etc.

Compute regime: batch size, grad accumulation, mixed precision rules.

Outside (excluded / treated exogenous):

Hardware faults / preemption events

Data pipeline outages

External evaluation gating not declared below

Boundary sanity checklist:

Data mixture is fixed or fully logged

Eval set and metric definitions are fixed or versioned

Any mid-run restart/resume is logged as a Switch event

A3) Δ — Timebase (what is one tick)

Tick unit Δ: (step / optimizer update / epoch / wall-clock window)
Logging cadence: every _____ ticks
Windowing for Ξ̂: W_k = [t_k, t_k + L] with L = _____ ticks
Smoothing (if any): _____ (must be declared)

A4) h — Observation / Compression Map

(A.2) z[n] := h(x(to + nΔ))

What is logged per tick/window (choose what you can actually guarantee):

Core training scalars

train loss

eval loss

eval accuracy / task score

learning rate

weight decay / reg coefficients

grad norm (global)

parameter norm(s)

Checkpoint/weight observables (optional)

checkpoint frequency: every _____ ticks

layers logged: {_____}

matrices logged: {W_q, W_k, W_v, W_o, MLP_in, MLP_out, ...}

derived spectral stats: singular values, effective rank, top-k energy

Representation observables (optional)

fixed diagnostic batch ID: _____

hidden states logged: layers {**}, tokens {**}

derived similarity: CKA/cosine drift, activation covariance eigenspectrum

CWA observables (optional but recommended)

multi-path prediction agreement across heads/layers

seed variance estimates

covariance / correlation diagnostics between micro contributors

Jump / regime-change indicators

change-point detector output

KL-like step jump proxy (must define)

restart/resume flags

A5) u — Operator Channels and Allowed Pulses

(A.3) $u \in \{\text{Pump, Probe, Switch, Couple}\}$

For each channel, define **what counts as an intervention** and how it is logged.

Pump (P) — fitting drive controls

Examples (tick-logged):

learning rate increase/decrease

batch size increase

loss scaling changes

gradient clipping changes

Pump knobs: {_____}
Pulse definition: δu_P := (parameter changed, magnitude, duration)

Probe (Q) — measurement / diagnostics

Examples:

running additional eval metrics

adding linear probes on frozen snapshots

logging more layers / adding diagnostic batch

Probe knobs: {_____}
Backreaction test planned? yes/no (see Gate 2)

Switch (Sw) — discrete regime changes

Examples:

optimizer swap (AdamW→SGD)

schedule phase change

curriculum stage boundary

data mixture swap

restart/resume

Switch events (must be timestamped):

Sw#1: time _____ ; description _____

Sw#2: time _____ ; description _____

Couple (C) — coherence / closure controls

Examples:

consistency regularization strength

auxiliary losses enforcing agreement

tying weights / sharing parameters

dropout/attention routing constraints

Couple knobs: {_____}
Pulse definition: δu_C := (constraint changed, magnitude, duration)

A6) Declared $\hat{\Xi}$ Proxy Set (must choose one)

(A.4) $\hat{\Xi}(W_k) := (\hat{\rho}(W_k), \hat{\gamma}(W_k), \hat{\tau}(W_k))$

Proxy Set Choice: A / B / C / D (from Section 11)
Exact formulas / code references: (must be copy/paste reproducible)

$\hat{\rho}$ (structure mass / concentration)

$\hat{\rho}$:= _____ (e.g., top-k spectral energy, activation concentration, –description length)

$\hat{\gamma}$ (coherence / coupling)

$\hat{\gamma}$:= _____ (e.g., CKA stability, redundancy correlation, constraint penalty response)

$\hat{\tau}$ (agitation / timescale separation)

$\hat{\tau}$:= _____ (e.g., $\tau_{delay} = t_{gen} - t_{fit}$, churn metric, recovery-vs-switch metric)

A7) Harness Gates (declare thresholds before running)

Gate 0 — loop existence + boundary sanity

ε_B (boundary residual tolerance) = _____

excluded segments: {restarts, outages, …} = _____

Gate 1 — proxy stability

c_ρ, c_γ, c_τ (CV thresholds) = _____

minimum windows to accept: _____

Gate 2 — probe backreaction

θ_Ô (tolerance for probe-on/off divergence) = _____

planned probe-on/off schedule: _____

Gate 3 — control effectiveness

Δ_min (minimum improvement in ||δΞ||) = _____

planned operator pulse tests: _____

A8) Regime Transition Claim Format (mandatory schema)

If you claim "sudden understanding happened," you must fill:

Event ID: SU#_____
Time interval: [t_a, t_b]
Metric jump: G(t) from _____ to _____
Ξ̂ movement: ($\hat{\rho}$, $\hat{\gamma}$, $\hat{\tau}$) from _____ to _____
Critical index: GCI(t) := $\kappa(P)\cdot\hat{\rho}\cdot\hat{\gamma}/(\hat{\tau}+\varepsilon_0)$
Threshold: Θ(P) = _____
Crossing evidence: GCI crosses Θ within [t_a, t_b]
Operator attribution: dominant channel(s) from logs u(t) = _____
Gate status: Gate0 pass? Gate1 pass? Gate2 pass? Gate3 pass? (must be yes for all)

A9) Minimal Logging Schema (JSON/YAML skeleton)

```
protocol_id: "P-____"
version: "v____"
boundary:
  model: {arch: "____", params: "____", depth: "____", width: "____"}
  optimizer: {type: "____", hyper: "____"}
  data: {dataset: "____", mix: "____", curriculum: "____"}
  regularizers: {weight_decay: "____", dropout: "____", others: "____"}
timebase:
  delta_unit: "step"
  log_every: ____
  window_length: ____
observation_map:
  scalars: [train_loss, eval_loss, eval_acc, lr, wd, grad_norm, param_norm]
```

```
  checkpoints: {enabled: false, every: null, layers: []}
  activations: {enabled: false, batch_id: null, layers: []}
operators:
  pump_knobs: ["lr", "batch_size", "clip"]
  probe_knobs: ["extra_eval", "logging_layers"]
  switch_events: []
  couple_knobs: ["consistency_weight", "tying"]
proxies:
  set: "A"
  rho_hat: "____"
  gamma_hat: "____"
  tau_hat: "____"
gates:
  gate0: {eps_boundary: ____}
  gate1: {cv_rho: ____, cv_gamma: ____, cv_tau: ____}
  gate2: {theta_backreaction: ____}
  gate3: {delta_min: ____}
events: []
```

A10) One-line Protocol Summary (for paper abstracts / OSF)

(A.5) P = (B, Δ, h, u): B=**; Δ=**; h=_____; u={Pump,Probe,Switch,Couple} with pulses **; proxies Ξ̂=**; gates {0..3} thresholds _____.

If you want, I can write **Appendix B (Ξ-Proxy Cookbook)** in the same copy/paste style, including concrete "loss-only," "checkpoint," and "activation" recipes plus a minimal Gate 2 probe-on/off test script outline (no code yet unless you ask).

Appendix B. Ξ-Proxy Cookbook (multiple proxy sets + pros/cons)

This cookbook lists **drop-in proxy sets** for $\hat{\Xi}(W) := (\hat{\rho}, \hat{\gamma}, \hat{\tau})$. Pick one based on what you can log, then **declare it in Protocol Card A** and validate with Gate 1 (stability) and Gate 2 (probe backreaction). ρ/γ/τ are **role-defined**; estimators are not privileged.

B0) Quick selection guide

Only scalars (loss/acc/schedule) logged → Proxy Set **S1**

Checkpoints available → Proxy Set **S2**

Hidden states on a fixed diagnostic batch available → Proxy Set **S3**

You can run operator pulses / probe-on-off tests → Proxy Set **S4** (best for causality)

Mixture-of-experts / many parallel paths → Proxy Set **S5** (CWA-focused)

S1) Scalar-only set (loss curves + schedules)

Use when: you cannot log weights/activations, only training/eval metrics.

ρ̂_S1 (structure mass)
(B.1) ρ̂ := z_eval_acc(W) (or −z_eval_loss(W), normalized)

γ̂_S1 (coherence)
(B.2) γ̂ := 1 − Gap(W)
where Gap can be train–test loss gap or any stable generalization gap metric.

τ̂_S1 (timescale separation / grokking delay)
(B.3) τ̂ := t_gen − t_fit

t_fit = first time train loss ≤ ε_fit

t_gen = first time eval score ≥ θ_gen

Pros

Works with minimal instrumentation.

Captures grokking-like delays directly (τ̂).

Cons

ρ̂ and γ̂ are entangled with evaluation choice; weak mechanistic specificity.

Gate 2 is hard: evaluation pipeline itself can backreact.

S2) Checkpoint spectral set (weights only)

Use when: you can save periodic checkpoints for selected layers/matrices.

Pick a small tracked set of matrices M = {Wℓ}. For each W, compute singular values s_i.

ρ̂_S2 (concentration of structure)
(B.4) ρ̂ := mean_{W∈M} (Σ_{i=1..k} s_i²)/(Σ_i s_i²)

Alternative (effective rank):
(B.5) ρ̂ := mean_{W∈M} 1/(r_eff(W)+ε0)
with r_eff = exp(−Σ p_i log p_i), p_i = s_i²/Σ s_j²

γ̂_S2 (lock-in / stability of top subspaces)
(B.6) γ̂ := mean_{W∈M} cos(span_topk(W(t)), span_topk(W(t−Δ)))

(Use principal angles between top-k singular vector subspaces.)

$\hat{\tau}$_S2 (churn)

(B.7) $\hat{\tau} := \text{mean}_{W \in M} ||W(t) - W(t-\Delta)||_F / (||W(t)||_F + \varepsilon_0)$

Pros

Much more mechanistic than S1; $\hat{\rho}$ becomes "concentration" not just "accuracy."

$\hat{\tau}$ is a true volatility/churn measure.

Cons

Requires checkpoint storage and SVD computation cost.

$\hat{\gamma}$ depends on stable top-k selection; can be noisy if spectra are flat (Gate 1 risk).

S3) Representation set (fixed diagnostic batch hidden states)

Use when: you can log hidden states for a fixed diagnostic batch D (must be versioned).

Let $H_\ell(t)$ be hidden states at layer ℓ on D.

$\hat{\rho}$_S3 (activation concentration)

(B.8) For each ℓ, let eigenvalues λ_i of $\text{Cov}(H_\ell)$.

$\hat{\rho} := \text{mean}_{\ell \in L} (\Sigma_{i=1..k} \lambda_i)/(\Sigma_i \lambda_i)$

$\hat{\gamma}$_S3 (coherence via representation stability + redundancy)

(B.9) $\hat{\gamma} := 0.5 \cdot \text{mean}_{\ell \in L} \text{CKA}(H_\ell(t), H_\ell(t-\Delta)) + 0.5 \cdot \text{Redundancy}(D)$

Where Redundancy(D) can be mean pairwise correlation of predictions from probes attached to multiple layers/heads.

$\hat{\tau}$_S3 (representation drift)

(B.10) $\hat{\tau} := \text{mean}_{\ell \in L} (1 - \text{CKA}(H_\ell(t), H_\ell(t-\Delta)))$

Pros

Directly targets "coupling/coherence" role of γ.

Often stable and informative for "lock-in" vs "churn."

Cons

Logging activations is expensive; privacy/security constraints may apply.

Requires careful Gate 2: diagnostic batch selection must not leak into training.

S4) Operator-lab set (pulse-based, PORE-native)

Use when: you can run small controlled interventions (operator pulses) and want causal attribution.

You keep one proxy family for each coordinate (choose from S2/S3), then add a **recovery-vs-switch** τ proxy and a gain test.

ρ̂_S4 = choose ρ̂_spec or ρ̂_act from S2/S3
γ̂_S4 = choose γ̂_CKA/redundancy from S3 or subspace stability from S2

τ̂_S4 (recovery-vs-switch)
(B.11) $\hat{\tau} := \max(T_{rec}, 1/(\hat{r}_J + \varepsilon_0))$

T_rec = time to return within ε of baseline Ξ̂ after a tiny Pump/Couple pulse

r̂_J = detected jump rate (change-point/KL-like indicator)

Add-on: local gain estimation (operator signatures)
(B.12) $\delta\Xi_{t+1} = \tilde{A}\,\delta\Xi_t + \hat{G}\,\delta u_t + \xi_t$

Pros

Strongest falsifiability: Gate 3 is meaningful.

Separates "what moved the system" by estimated gain vectors.

Cons

Requires experimental control; not always possible in large-scale production.

Needs careful jump rejection; cannot fit gains across Switch boundaries.

S5) CWA-focused set (many parallel contributors)

Use when: the architecture has many "contributors" (heads, experts, paths), and you care about macro cancellation.

Define micro contributors v_i as partial logits or partial predictions from each head/expert (or groups).

ρ̂_S5 (effective contributor count)
(B.13) $\hat{\rho} := 1 / (\Sigma_i p_i^2 + \varepsilon_0)$
where p_i = |v_i| / Σ_j |v_j| (a "participation ratio" / IPR-inverse)

γ̂_S5 (decorrelated diversity with stable macro)
(B.14) $\hat{\gamma} := 1 - \text{mean}_{i<j}\,\text{corr}(v_i, v_j)$ (bounded/clipped)

τ̂_S5 (macro stability lag)
(B.15) τ̂ := lag between "micro stabilization" and "macro stabilization"
e.g., t_macro − t_micro where t_micro is when v_i variance stabilizes, t_macro is when eval generalization stabilizes.

Pros

Directly operationalizes the CWA variance story (Var(Y) vs covariance terms).

Explains when “more width” helps vs hurts (covariance dominance).

Cons

Requires access to per-component outputs.

corr(v_i,v_j) depends on chosen decomposition; needs careful definition.

B6) Pros/cons summary table

Proxy set	Logging needed	Best for	Weakness
S1 Scalar-only	loss/acc/ schedule	cheap detection of delays ($\hat{\tau}$)	low mechanistic specificity; eval confounds
S2 Checkpoint spectral	weights	$\hat{\rho}$ as true concentration; $\hat{\tau}$ as churn	compute/storage; $\hat{\gamma}$ can be noisy
S3 Representation	activations on fixed batch	$\hat{\gamma}$ as lock-in; $\hat{\tau}$ as drift	expensive; batch leakage risk
S4 Operator-lab	pulses + (S2/S3)	causal operator attribution	needs controlled experiments
S5 CWA-focused	per-head/expert outputs	cancellation/diversity diagnosis	decomposition sensitive

---

B7) Minimal “cookbook rules” (avoid common mistakes)

Declare everything: diagnostic batch, layer list, k-values, thresholds.

Segment regimes: never compute smooth derivatives across Switch events.

Don’t conflate scale with structure: norm growth alone is not $\hat{\rho}$.

Gate 2 is real: if evaluation changes training decisions, probes backreact.

Prefer redundancy + stability over single numbers: $\hat{\gamma}$ should not be one fragile statistic.

If you want, next I can write **Appendix C (Toy Derivation Pack: from noisy voters to macro indicator)** in the same copy/paste style, keeping it short and Blogger-ready.

Appendix C. Toy Derivation Pack (mod-add): “micro noisy voters → macro stable decision”

This appendix is a **minimal derivation sketch** (not a full proof) of the modular-addition paper’s central mechanism: each neuron is a **noisy voter**, and the network becomes correct when a diversified population makes the noise **cancel** and the signal **add up**.

C1) Setup (the “toy universe”)

Let p be the modulus and x,y ∈ Z_p, with target class:

(C.1) s := (x + y) mod p

A two-layer model produces logits f[j] (j ∈ Z_p) by summing per-neuron contributions:

(C.2) f[j] := Σ_{m=1..M} f_m[j]

(Exact architectural details differ by activation choice; the paper’s mechanistic analysis is done in a setting where Fourier decomposition is clean.)

C2) Single-mode Fourier template (per neuron)

Empirically, each neuron’s weights concentrate on a single Fourier frequency k = φ(m), with cosine form:

(C.3) θ_m[t] = α_m·cos(ω_k·t + ϕ_m)
(C.4) ξ_m[t] = β_m·cos(ω_k·t + ψ_m)
(C.5) ω_k := 2πk/p

This is the paper’s “single-frequency Fourier feature” invariant.

C3) Intra-unit phase alignment (a key coupling condition)

A second invariant is a tight phase relationship (layer-wise coupling):

(C.6) (2ϕ_m − ψ_m) mod 2π = 0

This is the “alignment” that makes each neuron a coherent voter internally.

C4) Why one neuron is a *noisy* voter

Under (C.3)–(C.6), the paper shows each neuron’s contribution can be expanded into:

a **main term** that peaks at the correct class j = s, plus

structured residual terms that peak at “wrong but correlated” classes (notably j = 2x and j = 2y), plus

oscillatory cross-terms that depend on phases and frequency.

You can summarize this as:

(C.7) f_m[j] = Signal_m·1[j=s] + Noise_m,x·1[j=2x] + Noise_m,y·1[j=2y] + OscillatoryRemainder_m(j)

The important point is: **a single neuron does not implement a clean indicator**; it

implements a biased score with structured false positives.

C5) The cancellation trick: “full diversification” kills oscillatory noise in the sum

The paper defines a population condition (“full diversification”) consisting of:

balanced coverage across frequencies,

homogeneous scaling $\alpha_m \cdot \beta_m^2 = a$ (constant),

phase symmetry constraints within each frequency group.

Under (C.6) + diversification, the oscillatory remainders cancel when summed:

(C.8) $\Sigma_m \text{OscillatoryRemainder}_m(j) \approx 0$

so the network-level logit becomes a **closed-form flawed indicator** (paper’s Proposition 4.2 form):

(C.9) $f[j] = (aN/2)\cdot\{ -1 + (p/2)\cdot 1[j=s] \} + (p/4)\cdot(1[j=2x] + 1[j=2y])$

Here N is the number of neurons per frequency under diversification (so $M \approx N\cdot(\#\text{freqs})$).

Interpretation:

the first bracketed term is the **signal** and baseline,

the last term is the **structured residual noise** that survives cancellation.

C6) “Macro stable decision” = margin condition (why argmax/softmax becomes correct)

Let’s compare logits.

Correct class $j = s$:

(C.10) $f[s] = (aN/2)\cdot(-1 + p/2) + (p/4)\cdot(1[s=2x] + 1[s=2y])$

For generic inputs (where $s \neq 2x$ and $s \neq 2y$), the residual term is 0, so:

(C.11) $f[s] = aN\cdot(p-2)/4$

Typical wrong class $j \neq s$ and $j \neq 2x$ and $j \neq 2y$:

(C.12) $f[j] = -aN/2$

Worst structured wrong class (say $j = 2x$, similarly for $2y$):

(C.13) $f[2x] = -aN/2 + p/4$

Now the **margin** between the correct class and the worst structured wrong class is:

(C.14) $f[s] - f[2x] = aN\cdot(p-2)/4 - (-aN/2 + p/4)$
(C.15) $f[s] - f[2x] = (p/4)\cdot(aN - 1)$

So a minimal sufficient condition for correctness by argmax (and for softmax to concentrate on s) is:

(C.16) $aN > 1 \Rightarrow f[s] > f[2x]$ and $f[s] > f[2y]$ (generic case)

As aN increases above 1, the softmax probability on the correct class increases rapidly because the logit gap grows linearly in $p\cdot(aN-1)$. This is the **macro-stability threshold**: you do not need perfect micro voters; you need enough diversified voters so the **aggregate margin** becomes positive and then large.

C7) What this derivation is "for" in the integrated paper

This toy derivation supplies the concrete bridge used in Sections 5–6–9:

Micro: neurons are noisy voters; intra-unit alignment helps each voter become coherent.

Macro: cross-unit agreement is not required; diversification + symmetry yields cancellation and a stable decision.

Suddenness: once the effective margin crosses zero (the analog of crossing $\Sigma_c(P)$), observed generalization can look sharp even if internal change is gradual.

This is the cleanest "worked example" of the general pattern: **collapse-by-selection + stability-by-aggregation**.

Appendix D. CWA Taxonomy of Alignment (definitions + examples)

This appendix disambiguates "alignment," because the phrase **Collapse Without Alignment (CWA)** is otherwise easy to misread. The modular-addition laboratory *forces* this distinction: it requires **intra-unit** phase alignment, while simultaneously benefiting from **cross-unit non-alignment / diversity** plus symmetry cancellation.

We define three alignment types:

A_1: intra-unit alignment (within a single micro contributor)

A_2: cross-unit alignment (between contributors)

A_3: basis alignment (alignment to a privileged coordinate system / eigenbasis)

CWA (as used in CAFT) is fundamentally a statement about **A_2** (no need for cross-unit alignment), not about A_1 or A_3.

D1) A_1 — Intra-unit alignment (within-unit coherence)

Definition (A_1): A micro unit i is intra-aligned if its internal subcomponents are coordinated so that it produces a consistent contribution v_i to the macro observable.

Operationally: intra-unit alignment means the unit's own "encoding" and "readout" parts are phase-consistent, constraint-consistent, or self-coherent.

Canonical example (modular addition):
The paper observes and analyzes a layer-wise phase relation:

(D.1) $(2\phi_m - \psi_m) \bmod 2\pi = 0$

This is alignment **inside** neuron m between its input-side phase ϕ_m and output-side phase ψ_m. Without this, the neuron's output is less coherent as a voter.

Other examples (LLM context):

A single attention head whose Q/K/V subspaces stabilize so the head reliably implements one routing pattern.

A small circuit whose intermediate representation becomes consistent across contexts (reduced internal contradiction).

CWA stance on A_1: CWA does **not** forbid A_1; many CWA-stable macros actually benefit from units becoming internally coherent.

D2) A_2 — Cross-unit alignment (between-unit agreement)

Definition (A_2): A set of micro units $\{v_i\}$ are cross-aligned if they share a common internal phase/feature orientation, i.e., their contributions are mutually coherent in a way that reduces diversity (often increasing covariance).

Operationally: cross-unit alignment means the micro contributors "agree" in representation space, not just in output label.

Canonical counterexample (modular addition):
The modular-addition mechanism benefits from **diversification**: frequency coverage + phase symmetry across neurons. Neurons are not required to share one common phase; instead, phases are spread/symmetric so oscillatory noise cancels in the sum.

This is the crucial point:

A_2 is *not required* for macro correctness in that setting.

In some regimes, too much A_2 would **hurt** because covariance terms in Var(Y) would rise and cancellation would weaken (Section 6 logic).

Other examples (LLM context):

Many heads learning near-identical attention patterns (high redundancy). This can be good (robustness) or bad (wasted capacity) depending on whether redundancy improves macro stability or just increases correlated noise.

MoE experts collapsing to the same behavior (loss of specialization), which may reduce CWA-like cancellation benefits.

CWA stance on A2: CWA explicitly claims macro stability can arise **without** requiring A2, provided the macro observable is additivity-dominated and covariance is controlled.

D3) A3 — Basis alignment (alignment to a privileged eigenbasis)

Definition (A3): The system is basis-aligned if its learned structure concentrates in a coordinate system that diagonalizes the task's symmetries/invariances (an eigenbasis), making the solution sparse/clean in that basis.

Operationally: A3 is about whether the system has discovered "the right coordinates" for the task.

Canonical example (modular addition):
Because the task lives on Z_p, Fourier modes form the natural basis; the paper reports neurons become **single-frequency Fourier features** with cosine form.

Other examples (LLM context):

For certain structured tasks, the "right basis" may correspond to syntax/semantic factorization, algorithmic subroutines, or invariants under permutation/translation-like symmetries in the input distribution.

In mechanistic interpretability terms: discovering a sparse circuit basis where most behavior is captured by a small set of components.

CWA stance on A3: CWA does not require A3 globally; macro stability can exist even if the basis is not clean, as long as additive cancellation yields stable outputs. However, strong A3 often makes CWA easier (reduces variance and correlated noise).

D4) Summary: what "Collapse Without Alignment" actually denies

CWA is best read as:

(D.2) CWA denies "A2 is necessary for macro stability."

It does **not** deny:

that A1 might be necessary for a unit to be a reliable contributor,

that A3 can exist and can make learning more efficient.

D5) Diagnostic cues (how to tell which alignment is happening)

Diagnosing A1 (intra-unit)

internal consistency metrics improve within a unit (e.g., phase mismatch decreases; internal routing stabilizes).

unit becomes a more predictable voter (lower conditional variance of v_i given input class).

Diagnosing A2 (cross-unit)

pairwise correlations corr(v_i, v_j) rise across units,

diversity metrics drop (effective contributor count shrinks),

covariance term in Var(Y) becomes significant (bad for CWA cancellation).

Diagnosing A3 (basis)

spectral concentration in a candidate basis increases (lower effective rank in that basis),

low-dimensional structure explains more variance of activations/weights,

representation becomes sparse/clean under a specific transform.

D6) Practical rule (for the main paper's interventions)

If your macro stability relies on CWA cancellation: encourage **A1** (units become coherent) while preventing excessive **A2** (units become copies), and aim—where possible—for improved **A3** (discovering the task's invariants).

This rule is exactly what modular addition demonstrates: intra-unit alignment + cross-unit diversification + basis structure.

Appendix E. Minimal Simulation Loop (conceptual; not claiming realism)

This appendix gives a **pseudo-loop** that reproduces the *shape* of "grokking-like" behavior using the microdynamic template from Section 5 and the drive/cleanup competition from Section 8. It is **not** claimed to be a realistic LLM simulator—only a conceptual dynamical sketch that produces:

delayed winner-take-most mode selection (collapse),

then a sharp macro jump when a threshold is crossed (regime switch).

E1) State variables and meanings

We simulate a single "unit" with K candidate modes:

$A_k(t) \geq 0$: amplitude of mode k

$D_k(t) \in (-\pi, \pi]$: mismatch of mode k

$\kappa(t) := \text{Drive}(t)/\text{Cleanup}(t)$: force ratio (Section 8)

We also define a macro observable:

Y(t) : macro “decision confidence” (toy SNR-like)

E2) Micro update equations (discrete-time form)

Let Δ be a small timestep.

Mismatch relaxation (alignment-like):
(E.1) $D_k \leftarrow \mathrm{wrap}_\pi(D_k - \Delta\cdot\mu\cdot\sin D_k)$

Amplitude growth (competition with decay):
(E.2) $A_k \leftarrow A_k \cdot \exp(\Delta\cdot(\lambda_k\cdot\cos D_k - \beta(t)))$

λ_k: intrinsic advantage of mode k (fixed or curriculum-dependent)

$\beta(t)$: cleanup pressure (decay/regularization)

Normalization (optional, keeps numbers bounded):
(E.3) $A_k \leftarrow A_k / (\Sigma_j A_j + \varepsilon_0)$

This normalization is not required by theory; it is purely to prevent overflow in a conceptual loop.

E3) Drive/cleanup schedule and κ(t)

Choose a schedule where early training is drive-dominant and later cleanup grows:

(E.4) Drive(t) := 1
(E.5) Cleanup(t) := β(t)
(E.6) κ(t) := 1/β(t)

A simple “grokking-like” schedule:

(E.7) $\beta(t) := \beta_{min} + (\beta_{max} - \beta_{min})\cdot\mathrm{sigmoid}((t - t_0)/s)$

Early: $\beta \approx \beta_{min} \Rightarrow \kappa$ large (memorization phase)

Late: β increases ⇒ κ decreases (cleanup/refinement phase)

E4) Winner identification and collapse diagnostic

Define winner index:

(E.8) $w(t) := \mathrm{argmax}_k A_k(t)$

Dominance ratio:

(E.9) $R(t) := A_w(t) / (\Sigma_{j\neq w} A_j(t) + \varepsilon_0)$

“Collapse event” (micro):

(E.10) Collapse(t) := 1[R(t) ≥ R*]

E5) Macro observable and regime switch (toy CWA threshold)

We create a macro confidence Y(t) that increases when:

one mode dominates (high R),

and mismatch of the dominant mode is small (good internal coherence),

and κ is not too large (cleanup is active enough to remove residual noise).

One minimal form:

(E.11) Y(t) := (log(1+R(t))) · (1 − |sin D_w(t)|) · (1/(1+κ(t)))

Then define macro "generalization regime" by threshold crossing:

(E.12) Generalize(t) := 1[Y(t) ≥ Θ]

This produces: long plateau (Y below Θ), then a sharp rise when collapse + cleanup align.

E6) Pseudocode (copy/paste)

```
Initialize:
  choose K modes
  for k in 1..K:
    A_k ← small random >0
    D_k ← uniform in (−π, π]
  set λ_k (e.g., λ_1 slightly larger than others)
  set β(t) schedule parameters (β_min, β_max, t0, s)
  set thresholds R*, Θ
  set μ, Δ

For t = 1..T:
  # schedule
  β ← β_min + (β_max−β_min)*sigmoid((t−t0)/s)
  κ ← 1/(β + ε0)

  # micro updates
  for k in 1..K:
    D_k ← wrapπ( D_k − Δ*μ*sin(D_k) )
    A_k ← A_k * exp( Δ*( λ_k*cos(D_k) − β ) )

  # optional normalization
```

```
S ← sum(A_k) + ε0
for k in 1..K:
  A_k ← A_k / S

# collapse diagnostics
w ← argmax(A_k)
R ← A_w / (sum_{j≠w} A_j + ε0)

# macro confidence
Y ← log(1+R) * (1 – abs(sin(D_w))) * (1/(1+κ))

# regime flag
Generalize ← (Y ≥ Θ)

log t, β, κ, {A_k}, {D_k}, w, R, Y, Generalize
```

E7) What behavior you should see (qualitative)

With a typical setting (one λ_k slightly larger, random initial D_k):

Phase I (κ large, β small):

A_k grow but competition is slow; mismatch decreases gradually; R rises slowly.

Y remains below Θ (macro still "doesn't understand").

Phase II (κ decreases as β rises):

non-winner modes are suppressed by β; winner's advantage becomes decisive; R accelerates.

mismatch for winner is already smaller ⇒ cos D_w increases ⇒ even faster growth.

Y rises sharply; crossing Θ looks "sudden."

Phase III (cleanup-dominant):

winner remains; D_w stabilizes near 0; Y saturates high.

This reproduces the "delay then snap" pattern without claiming the loop is realistic.

E8) How to "stress test" the toy loop

Increase β_max too early: snap happens earlier but can reduce final A_w if β overwhelms growth.

Make λ_k nearly equal: t_c becomes very long (explains "needs many examples").

Force strong cross-mode correlation (not modeled here): you can mimic by tying D_k or adding covariance terms; Y may fail to cross Θ (CWA fails).

Next is a **second-level macro layer** with M independent units (voters), each running its own (A_k, D_k), and then define Y_total := Σ v_i to show explicit √M cancellation behavior (the CWA SNR scaling) in the same pseudo-loop style.

Appendix E (extension 1). Second-level macro layer with **M independent units (voters)**

Below is a conceptual **two-level** pseudo-loop:

Level 1 (micro): each unit i has K competing modes (A_i,k, D_i,k) and undergoes collapse-by-competition.

Level 2 (macro): the system decision aggregates **M unit votes** $v_i(t)$ into a macro score Y_total(t), which exhibits **CWA-style √M stabilization** when units are weakly correlated.

E9) State variables (two-level)

For unit i ∈ {1..M}, mode k ∈ {1..K}:

(E.18) $A_i,k(t) \geq 0$ (mode amplitude)
(E.19) $D_i,k(t) \in (-\pi, \pi]$ (mode mismatch)
(E.20) $w_i(t) := \text{argmax_k } A_i,k(t)$ (winner mode)
(E.21) $R_i(t) := A_i,w_i(t) / (\Sigma_\{j \neq w_i\} A_i,j(t) + \varepsilon_0)$ (dominance ratio)

Global:

(E.22) β(t) (cleanup schedule)
(E.23) κ(t) := Drive(t)/Cleanup(t) = $1/(\beta(t)+\varepsilon_0)$

E10) Level-1 micro update (per unit, per mode)

Discrete time step Δ:

Mismatch relaxation
(E.24) $D_i,k \leftarrow \text{wrap}\pi(D_i,k - \Delta \cdot \mu \cdot \sin D_i,k)$

Amplitude growth with decay
(E.25) $A_i,k \leftarrow A_i,k \cdot \exp(\Delta \cdot (\lambda_i,k \cdot \cos D_i,k - \beta(t)))$

Optional within-unit normalization (keeps amplitudes bounded)
(E.26) $A_i,k \leftarrow A_i,k / (\Sigma_j A_i,j + \varepsilon_0)$

E11) From micro state to a unit vote $v_i(t)$

We map each unit's internal "collapsedness + coherence + cleanup readiness" into a scalar vote.

Unit confidence

(E.27) $C_i(t) := \log(1+R_i(t)) \cdot (1 - |\sin D_i,w_i(t)|) \cdot (1/(1+\kappa(t)))$

Then define a **signed vote** toward "generalize" vs "not-yet":

(E.28) $v_i(t) := C_i(t) - b + \eta_i(t)$

b is a baseline offset (tunes difficulty)

$\eta_i(t)$ is unit noise (mean 0), conceptually capturing residual "wrong-but-correlated" behavior

You can also cap it to avoid extreme votes:

(E.29) $v_i(t) \leftarrow \text{clip}(v_i(t), -v_max, +v_max)$

E12) Level-2 macro aggregation (CWA layer)

Additive macro score:

(E.30) $Y_total(t) := \Sigma_{i=1..M} v_i(t)$

Regime flag:

(E.31) $Generalize_total(t) := 1[\ Y_total(t) \geq \Theta_M\ ]$

CWA stabilization (independence idealization):
If v_i are weakly correlated and share mean μ_v and variance σ_v^2:

(E.32) $E[Y_total] = M \cdot \mu_v$
(E.33) $Var(Y_total) \approx M \cdot \sigma_v^2$
(E.34) $SNR(Y_total) \approx \sqrt{M} \cdot (|\mu_v|/\sigma_v)$

So as M grows, the macro variable crosses the threshold more reliably and can look "sudden" once enough units become individually biased-positive.

E13) Optional: add cross-unit correlation (to test CWA failure)

To model correlation, set:

(E.35) $\eta_i(t) := \sqrt{c} \cdot \zeta(t) + \sqrt{(1-c)} \cdot \varepsilon_i(t)$

$\zeta(t)$ is a shared noise source, ε_i are independent, $c \in [0,1]$ is correlation strength.

Then:

(E.36) $Var(Y_total) = M \cdot \sigma^2 + M(M-1) \cdot Cov(v_i, v_j)$

As c increases, covariance dominates and $\sqrt{M}$ gains weaken—this is how you can simulate **CWA breakdown**.

E14) Two-level pseudocode (copy/paste)

```
Initialize:
  choose M units, K modes
  for each unit i:
    for k in 1..K:
      A[i,k] ← small random >0
      D[i,k] ← uniform(−π, π]
      λ[i,k] ← base advantage (option: one k has slight edge)
  set schedule β(t) (β_min, β_max, t0, s)
  set μ, Δ, baseline b, thresholds R*, Θ_M
  set noise model η_i(t) (independent or correlated)

For t = 1..T:
  β ← β_min + (β_max−β_min)*sigmoid((t−t0)/s)
  κ ← 1/(β + ε0)

  # Level 1: update all units
  for i in 1..M:
    for k in 1..K:
      D[i,k] ← wrapπ( D[i,k] − Δ*μ*sin(D[i,k]) )
      A[i,k] ← A[i,k] * exp( Δ*( λ[i,k]*cos(D[i,k]) − β ) )

    # optional normalize within unit
    S ← sum_k A[i,k] + ε0
    for k in 1..K:
      A[i,k] ← A[i,k] / S

    # winner and dominance
    w ← argmax_k A[i,k]
    R ← A[i,w] / (sum_{j≠w} A[i,j] + ε0)

    # unit confidence and vote
    C ← log(1+R) * (1 − abs(sin(D[i,w]))) * (1/(1+κ))
    η ← noise(i,t)   # independent or correlated
    v[i] ← clip(C − b + η, −vmax, vmax)

  # Level 2: macro aggregation
  Y_total ← sum_i v[i]
  Generalize_total ← (Y_total ≥ Θ_M)
```

```
log t, β, κ, summary stats of {R_i}, {C_i}, Y_total, Generalize_total
```

E15) What you should see (qualitative)

For small M, Y_total is noisy; even if some units collapse, macro may not cross Θ_M.

As M increases, macro becomes stable: once a fraction of units become positive-biased voters ($v_i > 0$), **Y_total crosses Θ_M sharply**.

If you turn on strong cross-unit correlation (large c), the √M benefit disappears; you can get "wide but not stable," matching the CWA warning.

Next is a tiny "diagnostic panel" definition (no code) for this simulation: e.g., plotting median $R_i(t)$, fraction collapsed, and Y_total(t), so the three-phase grokking structure becomes visible in the two-level model.

Appendix E (extension 2). Tiny diagnostic panel definition (no code)

This "panel" is a **minimal set of logged diagnostics** that makes the two-level simulation interpretable. Each item is defined as a scalar time series you compute at every tick t from the state $\{A_i,k(t), D_i,k(t)\}$ and votes $v_i(t)$. You can paste these definitions into your Protocol Card "h" section.

P1) Micro collapse dashboard (within-unit)

(E.40) Winner dominance (median):
R_med(t) := median_i $R_i(t)$

(E.41) Winner dominance (quantiles):
R_q10(t), R_q90(t) := 10th/90th percentile over i of $R_i(t)$

(E.42) Fraction collapsed:
FracCollapse(t) := $(1/M)\cdot|\{ i : R_i(t) \geq R^* \}|$

Interpretation:

Slow rise in R_med then rapid rise in FracCollapse signals the "collapse-by-competition" phase transition.

P2) Intra-unit coherence dashboard

(E.43) Winner mismatch (mean):
MisMean(t) := $(1/M)\cdot\Sigma_i |\sin(D_i,w_i(t))|$

(E.44) Winner mismatch (median):
MisMed(t) := median_i $|\sin(D_i,w_i(t))|$

Interpretation:

Falling MisMean/MisMed indicates increasing intra-unit alignment (A_1).

If FracCollapse rises but mismatch stays high, units are “dominant but incoherent” and votes will be noisy.

P3) Drive–Cleanup dashboard

(E.45) Cleanup pressure:
$\beta(t)$ (as scheduled)

(E.46) Force ratio:
$\kappa(t) := 1/(\beta(t)+\varepsilon_0)$

Interpretation:

Use $\kappa(t)$ to segment phases: κ high (memorization), κ mid (transition), κ low (refinement).

P4) Unit vote dashboard (micro → macro link)

(E.47) Vote mean and variance:
$\bar{v}(t) := (1/M)\cdot\Sigma_i\, v_i(t)$
$\mathrm{Var_v}(t) := (1/M)\cdot\Sigma_i\, (v_i(t) - \bar{v}(t))^2$

(E.48) Fraction positive voters:
$\mathrm{FracPos}(t) := (1/M)\cdot|\{\, i : v_i(t) \geq 0 \,\}|$

Interpretation:

Macro crossing typically occurs when FracPos(t) increases past a threshold and Var_v(t) is not exploding.

P5) CWA correlation dashboard (optional but very informative)

If you implemented correlated noise $\eta_i(t)$ with parameter c:

(E.49) Average pairwise vote correlation (sampled):
$\bar{\mathrm{Corr}}(t) := \mathrm{mean}_{(i,j)\ \mathrm{in}\ S}\ \mathrm{corr}(v_i(t-L..t), v_j(t-L..t))$

(where S is a sampled set of pairs, and corr is computed over a short window of length L.)

Interpretation:

If $\bar{\mathrm{Corr}}(t)$ rises, CWA cancellation weakens; macro stability may not improve with larger M.

P6) Macro stability dashboard (the “regime flag”)

(E.50) Macro score:
$Y_{total}(t) := \Sigma_i v_i(t)$

(E.51) Macro regime flag:
$Gen_{total}(t) := 1[\ Y_{total}(t) \geq \Theta_M\ ]$

(E.52) Normalized macro SNR proxy:
$\widehat{SNR}(t) := (M\cdot|\bar{v}(t)|) / \sqrt{(M\cdot Var_v(t) + \varepsilon_0)} = \sqrt{M}\cdot(|\bar{v}(t)|/\sqrt{(Var_v(t)+\varepsilon_0)})$

Interpretation:

The "sudden" moment is when $\widehat{SNR}(t)$ crosses an implicit threshold, often coincident with Gen_{total} flipping to 1.

P7) Minimal "phase segmentation" rules (for labeling plots)

You can label phases without extra machinery:

Phase I (memorization): $\kappa(t) \geq \kappa_{hi}$ and $Gen_{total} = 0$

Phase II (transition): $\kappa_{lo} < \kappa(t) < \kappa_{hi}$ and FracCollapse rising steeply

Phase III (refinement): $\kappa(t) \leq \kappa_{lo}$ and $Gen_{total} = 1$ with increasing $\widehat{SNR}$

(Choose κ_{hi}, κ_{lo} as protocol constants; keep them in the protocol card.)

Next is a tiny add-on: define a **single composite panel index** (like a "GCI_sim(t)") that mirrors Section 9's $\kappa\cdot\rho\cdot\gamma/\tau$ threshold form using only these diagnostics.

Appendix E (extension 3). Single composite panel index **GCI_sim(t)** (mirrors $\kappa\cdot\rho\cdot\gamma/\tau$)

We define a **simulation-only composite index** built strictly from the diagnostics in the panel (no extra hidden variables). It is designed to mirror:

(9.1) $\kappa(P)\cdot\rho(t)\cdot\gamma(t) / \tau(t) \geq \Theta(P)$

Step 1) Map diagnostics → proxy roles (ρ_{sim}, γ_{sim}, τ_{sim})

Structure mass / concentration (ρ_{sim}): use collapsed fraction (how many units have a dominant winner)

(E.60) $\rho_{sim}(t) := FracCollapse(t)$

Coherence / coupling (γ_{sim}): use winner mismatch complement (low mismatch = high coherence)

(E.61) $\gamma_{sim}(t) := 1 - MisMed(t)$
(where $MisMed(t) = median_i\ |\sin(D_i, w_i(t))|$)

Agitation / dephasing (τ_sim): use vote variance inflated by correlation (CWA-breaker)

If $\bar{Corr}(t)$ is available:

(E.62a) $\tau_sim(t) := \sqrt{(Var_v(t) + \varepsilon_0)} \cdot (1 + \alpha \cdot \max(0, \bar{Corr}(t)))$

If $\bar{Corr}(t)$ is not available:

(E.62b) $\tau_sim(t) := \sqrt{(Var_v(t) + \varepsilon_0)}$

Here α is a tunable sensitivity (set α=1 as default).

Step 2) Include force ratio κ(t)

From the panel:

(E.63) $\kappa(t) := 1/(\beta(t)+\varepsilon_0)$

Step 3) Define the composite index

(E.64) $GCI_sim(t) := \kappa(t) \cdot \rho_sim(t) \cdot \gamma_sim(t) / (\tau_sim(t) + \varepsilon_0)$

Interpretation

Increasing **FracCollapse** raises ρ_sim (more units have a dominant mode).

Decreasing **MisMed** raises γ_sim (units are internally coherent).

Increasing **Var_v** or **$\bar{Corr}$** raises τ_sim (macro agitation / CWA degradation).

Lower κ (more cleanup) will reduce GCI_sim if you use κ=Drive/Cleanup literally; if you prefer the Section 9 intuition "cleanup enables generalization," you can instead use the *inverse*:

(E.65) $\tilde{\kappa}(t) := 1/(1+\kappa(t)) = \beta(t)/(1+\beta(t))$
(E.66) $GCI_sim(t) := \tilde{\kappa}(t) \cdot \rho_sim(t) \cdot \gamma_sim(t) / (\tau_sim(t) + \varepsilon_0)$

Which variant to use?

Use **(E.64)** if you want to treat κ as a "drive dominance" indicator.

Use **(E.66)** if you want the index to increase when cleanup strengthens (often better for grokking-style plots).

Threshold rule (simulation regime flag)

(E.67) $Gen_total(t) \approx 1[\, GCI_sim(t) \geq \Theta_sim \,]$

Choose Θ_sim so the flip aligns with the macro flag (E.51) in a baseline run, then reuse Θ_sim for ablations.

This gives you a single curve that "predicts the jump" from panel-only observables while preserving the same structural form as Section 9's critical-surface condition.

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.